Uniform Bessel bounds for magnetization crossings in finite Ising chains

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Abstract

In a finite zero-field Ising chain, the all-up configuration and zero magnetization (magnetization one for an odd number of spins) become equally likely at a single coupling. This crossing has recently been described as a thermal transition of the finite chain, an analogue of a first-order transition temperature, and estimated by an approximate formula and a numerical fit. We prove its asymptotics, with explicit constants, for free and for periodic ends. The coupling at the crossing grows like half the logarithm of the length of the chain, minus half the logarithm of that logarithm, and periodic ends lower it by an amount that decays like the inverse logarithm of the length. The earlier approximations lack the doubly logarithmic term, which explains why a fit on chains of moderate length finds a slope below one half. At the crossing the central magnetization carries about as many domain walls as the logarithm of the length, so the crossing sharpens only like the inverse logarithm of the length, not like the inverse length as at a first-order transition. The proofs compare the central probability with a modified Bessel function, with a relative error bound that is explicit for every length, and for free ends they trap the crossing in an explicit interval at every length.

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1 Introduction

1.1 The crossing

Take $N \geq 2$ spins $\sigma_i = \pm 1$ with free ends and zero-field Ising weight $\exp(J \sum_{i<N} \sigma_i \sigma_{i+1})$, where $J > 0$. Here J is dimensionless, and it is the product βJ of the inverse temperature and the exchange coupling in the notation of [1]. The all-up configuration is the only one with magnetization $\sigma_1 + \cdots + \sigma_N = N$. Zero magnetization (or magnetization 1 for odd N, and the same holds for -1) comes from many configurations. For small J domain walls cost little and a central magnetization is the more likely of the two, and for large J the all-up configuration is. They are equally likely at a single coupling, the *central crossing*. Stepanyan et al. [1] study this crossing in the periodic chain as a thermal transition of a finite system, the analogue of the temperature of a first-order transition, and approximate it by $\beta J \approx (2 \log N - 1)/5$. Like them, we compare one central bin with the single all-up configuration (Remark 5.6 says what changes for other comparisons). We find its asymptotics with explicit constants for free and for periodic ends, and for free ends we bound it at every N.

Read from left to right, the spins are the steps of a persistent random walk [2, 3], a correspondence that García-Pelayo [4] uses to derive the exact magnetization laws of the open and the closed chain, also in a field. The first step is fair, and each later step repeats the previous one with probability p, where $e^{-2J} = (1-p)/p$. The magnetization is the position after N steps, so the walk that never turns, which reaches an end, competes with the many walks that end in the middle.

With two steps the walk ends in the middle exactly when it turns once, so the three bins have probabilities $\frac{p}{2}$, $1-p$ and $\frac{p}{2}$, and they are level at $p = \frac{2}{3}$. With three steps the bins have probabilities $\frac{p^2}{2}$, $\frac{1-p^2}{2}$, $\frac{1-p^2}{2}$ and $\frac{p^2}{2}$, and they are level at $p = 1/\sqrt{2}$. In general exactly one word reaches a given endpoint, namely the word that never turns, and very many words reach the middle by turning a few times (each turn brings a factor of about $1-p$).

The middle bin and the endpoints are level when the walk turns about $\log N$ times on average. More precisely, the crossing value p_N of p satisfies $N(1-p_N) = \log N - \frac{1}{2}\log\log N - 0.467\ldots + o(1)$, and for every $N \geq 2$ we trap $N(1-p_N)$ between two explicit functions of N whose distance tends to 0 (Theorem 3.4). For $p < p_N$ the middle is more likely, and for $p > p_N$ the ends are. In terms of the coupling, the crossing is at $J = \frac{1}{2}(\log N - \log\log N) + O(\log\log N/\log N)$ for free and for periodic ends, and the correlation length there is about $N/(2\log N)$ (Corollary 5.5). The approximations of [1] have no $\log\log N$ term, and this term explains why the slope of their numerical fit is below $\frac{1}{2}$ (Table 2). At the crossing the central bin carries about $\log N$ domain walls, so its energy exceeds that of the all-up configuration only by about $2\log N$, and the crossing sharpens like $1/\log N$ in the coupling, not like $1/N$ as at a first-order transition (Proposition 5.7).

The crossing lies between two regimes in which the law is well understood. When $p = 1-\tau/N$ with $\tau > 0$ fixed, the rescaled walk tends to the integrated telegraph process and the magnetization per spin has a scaling limit [5, Theorem 2.1 and Remark 3.4]. When $0 < p < 1$ is fixed, the bin has a normal limit (Theorem 2.5). At the crossing τ grows like $\log N$, so neither limit decides it, and we need bounds that hold uniformly in τ. Our main tool is a comparison of the middle bin with a modified Bessel function, with a relative error that is explicit for every N (Theorem 3.3).

A walk with d directions, which is the d-state Potts chain, is treated in [6], and general slowly switching chains in [7].

1.2 Setting and notation

The walk. The walk makes $N \geq 1$ steps. We record a run as a word $w = w_1 \cdots w_N$ in the letters $+$ (right) and $-$ (left). The letter w_1 is $+$ or $-$ with probability $\frac{1}{2}$, and given $w_1, \ldots, w_i$,

$$\mathbb{P}(w_{i+1} = w_i) = p, \qquad \mathbb{P}(w_{i+1} \neq w_i) = q = 1-p.$$

An index $i < N$ with $w_i \neq w_{i+1}$ is a *switch*. Let $\mathrm{chg}(w)$ be the number of switches and $\mathrm{rep}(w) = N - 1 - \mathrm{chg}(w)$. Then

$$\mathbb{P}(w) = \tfrac{1}{2}\, p^{\mathrm{rep}(w)} q^{\mathrm{chg}(w)}. \tag{1}$$

For $p > 0$ we put $u = q/p$, the odds of a switch.

Bins. The walk ends in bin k if w has k letters $+$, for $0 \leq k \leq N$. We write $P_N(k)$ for the probability of bin k, and $P_0(0) = 1$. Bins 0 and N are the *endpoints*. Only the constant word $-\cdots-$ reaches bin 0, and only $+\cdots+$ reaches bin N, so

$$P_N(0) = P_N(N) = \tfrac{1}{2}p^{N-1} \qquad (N \geq 1).$$

A middle bin is $m_N = \lfloor N/2 \rfloor$. For odd N the two middle bins have equal probability by symmetry.

Spins. Put $\sigma_i = +1$ if $w_i = +$ and $\sigma_i = -1$ if $w_i = -$, and $X_N = \sigma_1 + \cdots + \sigma_N$. The bin is $K_N = (N + X_N)/2$. Since $\mathrm{chg}(w) = \sum_{i<N}(1 - \sigma_i\sigma_{i+1})/2$, (1) gives $\mathbb{P}(w) \propto \exp(J\sum_{i<N}\sigma_i\sigma_{i+1})$ with $e^{-2J} = u$. So the spins form the Ising chain in zero field with free boundaries, and $J > 0$ exactly when $p > \frac{1}{2}$.

Scales. Let $L = \log N$ and $\tau = Nq$. Each of the $N - 1$ steps after the first switches with probability q, so the expected number of switches is $(N-1)q = \tau - q$. We call τ the *switching scale*. In the Ising chain a switch is a domain wall, and $q = 1/(1 + e^{2J})$, so τ is the expected number of domain walls plus q. The central crossing has τ of order L. Near the edges the crossings have larger τ, up to order $\sqrt{N}$ for the bins next to an edge (Section 4).

Ratios and crossings. For a bin k with $1 \le k \le N - 1$ we write $a = k$ and $b = N - k$ for its numbers of letters $+$ and $-$. For $p > 0$ the ratio of bin k to an endpoint depends on p only through u, and it is a polynomial in u with nonnegative integer coefficients ((13) below). We write

$$S_{a,b}(u) = \frac{P_N(a)}{P_N(0)}, \qquad S_N(u) = S_{m_N,\,N-m_N}(u)$$

for this ratio and for its value at the middle. *The crossing* of bin k is the value of p at which bin k is exactly as likely as an endpoint, that is, $S_{a,b}(u) = 1$. It is unique (Section 4), and we write $p_{a,b}$ for it and $u_{a,b} = (1 - p_{a,b})/p_{a,b}$ for its odds. The *central crossing* is $p_N = p_{m_N,N-m_N}$, with odds u_N, and

$$z_N = N(1 - p_N) = \frac{Nu_N}{1 + u_N} \tag{2}$$

is the value of τ there.

Special functions and asymptotic notation. The functions I_0 and I_1 are the modified Bessel functions of the first kind, of orders zero and one. We use them through their power series and their behavior at infinity [8, Eqs. 10.25.2, 10.40.1],

$$I_0(2x) = \sum_{r\ge0}\frac{x^{2r}}{(r!)^2}, \quad I_1(2x) = \sum_{r\ge0}\frac{x^{2r+1}}{r!\,(r+1)!}, \quad I_\nu(z) = \frac{e^z}{\sqrt{2\pi z}}\Big(1 - \frac{4\nu^2-1}{8z} + O(z^{-2})\Big)$$

as $z \to +\infty$ along the positive real axis, for $\nu = 0, 1$. We write $f \sim g$ if $f/g \to 1$. Each $O(\cdot)$ comes with the parameters it is uniform in, and $O_a(\cdot)$ means that the constant may depend on a.

Ising dictionary. Sections 2 to 4 are written in the variables of the walk, and they translate as follows. Bin k is magnetization $X_N = 2k - N$, so the endpoints are full magnetization $\pm N$. A middle bin has magnetization zero for even N and either -1 or 1 for odd N. A switch is a domain wall, and its odds $u = e^{-2J}$ are the Boltzmann factor of one wall. The crossing of bin k is the coupling $J_{a,b} = -\frac{1}{2}\log u_{a,b}$ at which magnetization $a - b$ and magnetization N are equally likely, and at the center it is $J_N^{\mathrm{F}} = -\frac{1}{2}\log u_N$. The correlation length of the infinite chain in zero field, with no constraint on the magnetization, is $\xi(J) = -1/\log\tanh J$, and $\tanh J = (1-u)/(1+u)$, so $N/\xi = N\log\frac{1+u}{1-u} = 2Nu(1 + O(u^2))$. Since $\tau = Nu/(1+u)$, this gives $\tau = \frac{N}{2\xi}(1 + O(u))$ as $u \to 0$. To leading order, the switching scale is half the length of the chain measured in correlation lengths. It is exactly the scaling variable $\zeta = N(1 - \tanh J)/2$ of Antal, Droz and Rácz [9] (we keep the letter ζ for ζ_N below).

1.3 Main results

The coupling at the crossing. In Ising terms the free central crossing is the coupling $J_N^{\mathrm{F}} = -\frac{1}{2}\log u_N$, and J_N^{P} is the coupling at the periodic one. Put $c_0 = \frac{1}{2}\log(\pi/8) = -0.467\ldots$ and $c_{\mathrm{P}} = \frac{1}{2}\log(\pi/2)$. Theorem 5.2 and Corollary 5.5 give

$$J_N^X = \tfrac{1}{2}(\log N - \log\log N) + \frac{\frac{1}{4}\log\log N - \frac{1}{2}c_X}{\log N} + O\Big(\frac{(\log\log N)^2}{(\log N)^2}\Big)$$

for free ends ($X = \mathrm{F}$, with $c_{\mathrm{F}} = c_0$) and for periodic ends ($X = \mathrm{P}$). Since $c_{\mathrm{P}} = c_0 + \log 2$, periodic ends lower the coupling at the crossing by $(\log 2)/(2\log N)$ to leading order. At either crossing the correlation length is about $N/(2\log N)$, a vanishing fraction of the chain. The approximation $\beta J \approx (2\log N - 1)/5$ of [1, Eq. (16)] grows too slowly. At $N = 10^5$ the exact periodic coupling is 4.576, the periodic case of the display, without its error term, gives 4.578, and the approximation gives 4.405 (Table 2).

A Bessel bound at every N. Put $h = N/2$ and $F(x) = 2x(I_0(2x) + I_1(2x))$. Theorem 3.3 says that for every $N \geq 2$ and every $x > 0$,

$$0 \leq 1 - \frac{hS_N(x/h)}{F(x)} \leq \frac{x(x+1)}{h}.$$

Near the crossing $x = Nu/2$ is about $L/2$, so the relative error is $O(L^2/N)$ (tiny for large N). Theorem 4.3 gives the same kind of bound for every bin, with relative error at most $Nu^2/2 + u$.

The central crossing at every N. Let ζ_N be the positive root of $8\zeta e^{2\zeta} = \pi N^2$, that is, $\zeta_N = \frac{1}{2}W_0(\pi N^2/4)$ with W_0 the principal branch of Lambert's function, and let $\varepsilon_N = \zeta_N(\zeta_N + 1)/N + 1/(16\zeta_N^2)$. Theorem 3.4 says that

$$\zeta_N + \frac{1}{8\zeta_N} - \varepsilon_N < z_N < \zeta_N + \frac{1}{8\zeta_N} \qquad \text{for every } N \geq 2.$$

Note that $\varepsilon_N \to 0$. In terms of the coupling this traps J_N^{F} between $\frac{1}{2}\log(N/y - 1)$ at $y = \zeta_N + \frac{1}{8\zeta_N}$ and at $y = \zeta_N + \frac{1}{8\zeta_N} - \varepsilon_N$ (Corollary 3.7(a)). For $N \geq 175$ the proof is analytic. For $2 \leq N \leq 174$ we decide the sign of $S_N - 1$ at two rationals, one on each side of the root, in exact integer arithmetic, and the Lean kernel repeats this check. The theorem has two simpler consequences. With c_0 as above, the crossing satisfies

$$z_N + \tfrac{1}{2}\log z_N = L + c_0 + \frac{1}{8z_N} + R_N, \qquad |R_N| \leq \frac{1}{4L^2} + \frac{L^2 + L}{N} \quad (N \geq 2)$$

(Proposition 3.8), and as $N \to \infty$

$$z_N = L - \tfrac{1}{2}\log L + c_0 + \frac{\frac{1}{4}\log L - \frac{1}{2}c_0 + \frac{1}{8}}{L} + O\Big(\frac{(\log L)^2}{L^2}\Big)$$

(Corollary 3.7). The convergence is slow, and z_N/L is still only 0.84 at $N = 10^4$ (Figure 2).

Every bin. Let bin a be at distance a from the nearer edge, with $b = N - a \geq a$. Next to the edge the crossing is at $p_N^{\mathrm{adj}} = (1 + \sqrt{N-1})/(2 + \sqrt{N-1})$. For $N \geq 4$ the two bins next to the edges are the least likely of all at the central crossing (Proposition 4.2). As $a \to \infty$, Theorem 4.7 expands $2\sqrt{ab}\,u_{a,b}$ to the same order as z_N, uniformly in $b \geq a$. For fixed a the crossing comes instead from a Laguerre polynomial. Then $u_{a,b} = c_a/\sqrt{b} - 1/b + O_a(b^{-3/2})$, where c_a^2 is the positive root of an explicit polynomial of degree a (Theorem 4.8). The two regimes match (Theorem 4.9).

Periodic boundaries. If the steps are closed into a loop, so that the last step also interacts with the first, the central crossing moves by $(\log 2)/N$ in the odds to leading order (Theorem 5.2). Under either boundary, at odds $u_N^X + t/N$, where u_N^X is the central crossing of that boundary, the ratio of bin $N/2 + yN/\sqrt{\log N}$ to an endpoint tends to e^{t-2y^2} (Theorem 5.3).

Domain walls and the energy gap. At the free crossing the central bin carries $z_N + \frac{1}{2} + O(1/z_N)$ domain walls on average (Proposition 5.7), so its mean energy exceeds that of the all-up configuration by about $2\log N$ in units of the exchange constant. At a first-order transition the two coexisting phases differ in energy by an amount proportional to N. Accordingly the ratio of the central bin to an endpoint changes by a factor e over a range of couplings of width about $1/(2\log N)$. The $\log\log N$ term is the cost of comparing full order with one bin instead of the whole disordered band. The probability that the chain is fully ordered is $\frac{1}{2}$ at a coupling larger than J_N^{F} by $\frac{1}{2}\log\log N + O(1)$ (Remark 5.8).

We prove these results in Sections 2 to 5, in this order. Section 6 lists what was computed and what is in Lean, and the appendix has the longer proofs of Section 3.

1.4 Related work

Early treatments of the correlated random walk include Goldstein [2] and Gillis [3]. Renshaw and Henderson [10] study the symmetric correlated walk, derive exact transition probabilities and investigate limiting distributions. Konno [11, Lemma 2 and Theorem 7] writes the exact law with the hypergeometric function ${}_2F_1$ and proves the integrated telegraph limit when $p = 1 - \theta/N$ with $\theta \in (0, 1)$ fixed. Herrmann and Vallois [5, Theorem 2.1 and Remark 3.4] give the process limit and its law, including the Bessel density and endpoint atoms. With τ replaced by Nu, their law gives the Bessel form of every bin that we use in Section 4.2. These fixed-scale limits do not by themselves give the uniform finite-N bounds needed when τ grows like $\log N$. Dekking and Kong [12] study the Markov binomial law, give its variance and find when it has one, two or three modes. Their results already give the crossing of the bins next to the edges and the ordering of the bins in a row (Section 4.1).

For the Ising chain, Antal, Droz and Rácz [9] give the magnetization law at a fixed scale τ under several boundary conditions, and Dantchev and Rudnick [13] give exact free and periodic coefficients. Dantchev, Tonchev and Rudnick replace these coefficients by Bessel functions at a fixed value of their scaling variable $x_t = 2Nu$ (which is about 2τ), for periodic ends [14, Section 4.3.1 and Appendix D] and with a relative error $O(1/N)$ for free ends [15, Section 3, Eq. (3.10)]. These errors are stated for x_t of order one, while at the crossing x_t grows like $2\log N$. The scaling variable of Antal, Droz and Rácz is our τ, and their fixed-τ scaling law already predicts the central crossing if one evaluates it at τ of order $\log N$ (a regime in which it is not proved). In that law the magnetization per spin has a density, and neighboring bins are $2/N$ apart. So the middle bin and an endpoint balance when the endpoint atom $e^{-\tau}/2$ equals $2/N$ times the density at zero magnetization, which is $\frac{1}{2}\tau e^{-\tau}(I_0(\tau) + I_1(\tau))$ by [9, Eq. (47)]. This gives $\tau(I_0(\tau) + I_1(\tau)) = N/2$, which is the equation $F(x_B) = h$ of the Bessel root of Theorem 3.3(ii) with $\tau = 2x_B$. Its root z satisfies $z + \frac{1}{2}\log z = L + c_0 + 1/(8z) + O(z^{-2})$, the crossing equation of Proposition 3.8 without its remainder. So the scaling law gives the Bessel root. What we add is a proof that the Bessel root is the crossing up to a relative error $O(L/N)$ in the odds, with explicit bounds at every N, and the bracket of Theorem 3.4. In the same way the large-argument form of the known Bessel laws suggests the $\log\log N$ term, the periodic shift by $\log 2$ and the Gaussian profile of Theorem 5.3, and for these our contribution is the proof. Stepanyan et al. [1, Section III.B] study two crossings

of the periodic chain as thermal transitions. In the first, the all-up configuration and magnetization $N-2$ (one flipped spin) are equally likely exactly at $J=\frac{1}{4}\log N$, as they show. In the second, the all-up configuration and zero magnetization are equally likely, and they estimate the coupling. We give the second with its constants under both boundary conditions, and for free ends we bound it at every N and find the crossing of every bin. The Markov binomial estimates of Čekanavičius and Mikalauskas assume ranges of the transition probabilities that exclude $p\to 1$ in this symmetric model (see [16, Theorems 1.1–1.3, pp. 220–221] and [17, Theorems 1.1–1.3, pp. 308–309]).

In short, the exact laws, the Bessel forms and their fixed-scale limits are known, and we use them as inputs. What is new here, as far as we know, is quantitative. First, we prove relative Bessel bounds that are explicit in N, in u and in the bin (Theorems 3.3, 4.3 and 5.1), and that stay small when the mean number of domain walls grows like $\log N$. Second, we bracket the free crossing at every N (Theorem 3.4). Third, we expand the actual crossings, uniformly over $b\ge a$ near a growing edge (Theorem 4.7), and describe the crossing window about the exact root (Theorem 5.3).

2 The exact law

The exact law and recurrence are classical [12, Proposition 4.1 and Eq. (4.1)] (the law is also in [4], in terms of ${}_2F_1$), as are the generating function [3, Eq. (3.1)] and the fixed-p normal limit [11, Theorem 6]. We prove Theorem 2.1 because the later sections use its count of words by switches. By (1), a word with j switches has probability $\frac{1}{2}p^{N-1-j}q^j$.

Theorem 2.1 (bin probabilities)**.** *Let $N\ge 1$. Then $P_N(0)=P_N(N)=\frac{1}{2}p^{N-1}$, and for $1\le k\le N-1$*

$$P_N(k)=\frac{1}{2}\sum_{j=1}^{N-1}W(N,k,j)\,p^{N-1-j}q^j,$$

where $W(N,k,j)$ counts the words with k letters $+$, $N-k$ letters $-$ and exactly j switches. Writing $a=k$ and $b=N-k$, for every integer $i\ge 1$

$$W(N,k,2i-1)=2\binom{a-1}{i-1}\binom{b-1}{i-1},\qquad W(N,k,2i)=\binom{a-1}{i}\binom{b-1}{i-1}+\binom{a-1}{i-1}\binom{b-1}{i}$$

with $\binom{n}{r}=0$ for integers $n\ge 0$ and r outside $0\le r\le n$.

Proof. By (1) it suffices to count the words in bin k with j switches. We count them by their run lengths. A word with j switches consists of $j+1$ maximal runs, alternately of letters $+$ and of letters $-$. If $k\in\{0,N\}$ there is one word and it has no switch. Let $a,b\ge 1$. If $j+1=2i$ there are i runs of each letter. The run lengths form a composition of a into i parts and a composition of b into i parts, and there are $\binom{a-1}{i-1}\binom{b-1}{i-1}$ such pairs. The two choices of the first letter give the factor 2. If $j+1=2i+1$ the word starts and ends with the same letter. Starting with $+$ gives $i+1$ runs of $+$ and i runs of $-$, so $\binom{a-1}{i}\binom{b-1}{i-1}$ words, and starting with $-$ gives the other summand. □

For instance $P_2=(\frac{p}{2},\,q,\,\frac{p}{2})$, $P_3(1)=\frac{1}{2}(1-p^2)$ and $P_4(2)=q(1-p+p^2)$. Each interior bin needs at least one switch, while the endpoints need none. So the endpoints win when p is close to 1 (the walk rarely turns), and Section 3 asks how close.

Proposition 2.2 (recurrence and generating function)**.** *Let $P_N^+(k)$ and $P_N^-(k)$ be the probabilities of being in bin k after N steps with last letter $+$, respectively $-$, and set them to 0 outside $0 \le k \le N$. Then $P_1^+(1) = P_1^-(0) = \frac{1}{2}$, $P_1^+(0) = P_1^-(1) = 0$, and for $N \ge 2$*

$$P_N^+(k) = p\,P_{N-1}^+(k-1) + q\,P_{N-1}^-(k-1), \qquad P_N^-(k) = p\,P_{N-1}^-(k) + q\,P_{N-1}^+(k). \tag{3}$$

Consequently

$$\sum_{N\ge 0}\sum_{k=0}^{N} P_N(k)\,x^k t^N = \frac{1-(p-\frac{1}{2})(1+x)\,t}{1-p(1+x)\,t+(2p-1)\,x\,t^2}, \tag{4}$$

and for all $N \ge 2$ and all k, with $P_M(k) = 0$ outside $0 \le k \le M$,

$$P_N(k) = p\big(P_{N-1}(k) + P_{N-1}(k-1)\big) - (2p-1)\,P_{N-2}(k-1). \tag{5}$$

Proof. The recursion (3) follows by conditioning on w_{N-1}, since w_N equals w_{N-1} with probability p. Let $H^+ = \sum_{N,k} P_N^+(k) x^k t^N$ and define H^- alike. By (3), $H^+ = xt(\frac{1}{2} + pH^+ + qH^-)$ and $H^- = t(\frac{1}{2} + pH^- + qH^+)$. Solving these two linear equations and adding 1 for $N = 0$ gives (4), which is Gillis's generating function [3, Eq. (3.1)] after changing from position to bin number. Multiplying (4) by its denominator and comparing the coefficients of $x^k t^N$ with $N \ge 2$ gives (5). □

Remark 2.3 (integer numerators). For $p = \alpha/(\alpha+\beta)$ with integers $\alpha, \beta \ge 0$, not both 0, put $n_N(k) = 2(\alpha+\beta)^{N-1} P_N(k)$ for $N \ge 1$. By (1), $n_N(k)$ is the sum of $\alpha^{\mathrm{rep}(w)}\beta^{\mathrm{chg}(w)}$ over the words with k letters $+$ and $N-k$ letters $-$, which shows that it is an integer.

2.1 Variance and the normal limit

Note that K_N counts the visits of the chain $w_1, \dots, w_N$ to the letter $+$. This chain leaves each letter with probability q and starts from its stationary law $(\frac{1}{2}, \frac{1}{2})$. So K_N has the Markov binomial law $\mathrm{Bin}(N, q, q, (\frac{1}{2}, \frac{1}{2}))$ of Dekking and Kong [12], whose second eigenvalue is $1 - 2q = 2p - 1$.

Proposition 2.4 (mean and variance)**.** *Let $0 < p < 1$ and $\rho = 2p - 1$. Then $\mathbb{E}K_N = N/2$ and*

$$\operatorname{Var} K_N = \frac{1}{4}\Big(N\,\frac{1+\rho}{1-\rho} - \frac{2\rho\,(1-\rho^N)}{(1-\rho)^2}\Big) = \frac{Np}{4(1-p)} - \frac{(2p-1)\big(1-(2p-1)^N\big)}{8(1-p)^2}.$$

Proof. This is [12, Eq. (2.3) and Proposition 2.1] with $a = b = q$ and the stationary start, for which both excentricities vanish and both stationary weights are $\frac{1}{2}$. The second form follows from $(1+\rho)/(1-\rho) = p/q$ and $1 - \rho = 2q$. □

Theorem 2.5 (normal limit)**.** *Let $0 < p < 1$ be fixed. Then $(K_N - N/2)/\sqrt{N}$ converges in distribution, as $N \to \infty$, to the normal law with mean 0 and variance $p/(4(1-p))$.*

Proof. By (3), the vector $v_N(\theta)$ with entries $\mathbb{E}[e^{i\theta X_N}; \sigma_N = \pm 1]$ satisfies $v_{N+1} = M v_N$, where M has rows $(pe^{i\theta}, qe^{i\theta})$ and $(qe^{-i\theta}, pe^{-i\theta})$. Its eigenvalues at $\theta = 0$ are 1 and $2p-1$, so for small $|\theta|$ they stay simple, with $\lambda_+(\theta) = p\cos\theta + \sqrt{q^2 - p^2\sin^2\theta} = 1 - \frac{p}{2q}\theta^2 + O(\theta^4)$ and λ_- near $2p-1$. Since $v_1(0)$ is an eigenvector for the eigenvalue 1, the component along λ_+ tends to 1, and at $\theta = s/\sqrt{N}$ we get $\mathbb{E}e^{i\theta X_N} \to \exp(-\frac{p}{2q}s^2)$. Lévy's continuity theorem [18, Theorem 26.3] gives the normal limit of $X_N/\sqrt{N}$ with variance p/q, and $K_N - N/2 = X_N/2$. □

For $p > \frac{1}{2}$ persistence widens the bell by the factor $\sqrt{p/q}$ compared with the simple random walk, and for $p < \frac{1}{2}$ it narrows it by the same factor.

3 The central crossing at every N

Let $N \geq 2$, let $m_N = \lfloor N/2 \rfloor$ and let $p > 0$. By Theorem 2.1 the ratio of the middle bin to an endpoint is

$$\frac{P_N(m_N)}{P_N(0)} = S_N(u) := \sum_{j=1}^{N-1} W(N, m_N, j)\, u^j, \tag{6}$$

a polynomial in u with nonnegative integer coefficients and $W(N, m_N, 1) = 2$. The middle and the endpoints are level exactly when $S_N(u) = 1$.

We first give a heuristic for why this happens when τ is near $\log N$. Let $a = m_N$ and $b = N - m_N$, both about $N/2$. A word in the middle bin with $2r+1$ switches has $r+1$ runs of each letter, and there are $2\binom{a-1}{r}\binom{b-1}{r} \approx 2(N/2)^{2r}/(r!)^2$ of them when r is small compared with $\sqrt{N}$. So these words contribute about $2u\sum_r (Nu/2)^{2r}/(r!)^2 = 2uI_0(Nu)$ to $S_N(u)$, and in the same way the words with an even number of switches contribute about $2uI_1(Nu)$. For large $w = Nu$ we have $I_0(w) + I_1(w) \approx 2e^w/\sqrt{2\pi w}$, so $S_N(u) \approx 4we^w/(N\sqrt{2\pi w})$. Setting this equal to 1 gives $w + \frac{1}{2}\log w \approx L + \frac{1}{2}\log(\pi/8)$. So at the crossing w, and with it τ, is close to $L - \frac{1}{2}\log L + c_0$ with $c_0 = \frac{1}{2}\log(\pi/8)$.

Theorem 3.1. *(a) For every $N \geq 2$ there is exactly one $p = p_N \in [0,1]$ with $P_N(m_N) = P_N(0)$, and $p_N \geq \frac{2}{3}$. The middle is more likely than the endpoints for $p < p_N$ and less likely for $p > p_N$.*

(b) The first values are $p_2 = \frac{2}{3}$, $p_3 = 1/\sqrt{2}$ and the real root p_4 of $3p^3 - 4p^2 + 4p - 2$, and $p_N < p_{N+1}$ for all $N \geq 2$.

(c) As $N \to \infty$ we have $p_N \to 1$, and more precisely $1 - p_N \sim (\log N)/N$.

Proof. (a) At $p = 0$ the endpoints have probability 0 and the middle does not. For $p > 0$ the equation is $S_N(u) = 1$ by (6). The coefficients of S_N are nonnegative and the linear one is 2, so S_N is strictly increasing on $[0, \infty)$ with $S_N(0) = 0$ and $S_N(u) \geq 2u$. By the intermediate value theorem there is exactly one root $u_N \in (0, \frac{1}{2}]$. The map $u \mapsto p = 1/(1+u)$ is a decreasing bijection from $[0, \infty)$ to $(0, 1]$. Hence $p_N = 1/(1+u_N) \geq \frac{2}{3}$, and the middle wins exactly when $u > u_N$.

(b) By Theorem 2.1, $S_2 = 2u$, $S_3 = 2u + u^2$ and $S_4 = 2u + 2u^2 + 2u^3$, which give the three values. For p_4, substitute $u = (1-p)/p$ in $S_4(u) = 1$. For the monotonicity we compare coefficients. Going from N to $N+1$, one entry of the middle pair $(a, b) = (\lfloor N/2 \rfloor, \lceil N/2 \rceil)$ grows by one and the other is unchanged. The formulas of Theorem 2.1 are symmetric in (a, b), and every $\binom{a-1}{\cdot}$ is nondecreasing in a. So $W(N+1, m_{N+1}, j) \geq W(N, m_N, j)$ for every j. The coefficient of u^2 is $a + b - 2 = N - 2$, which increases strictly. Hence $S_{N+1}(u) > S_N(u)$ for $u > 0$, so $u_{N+1} < u_N$.

(c) We prove (c) after Theorem 3.3, from part (i) of that theorem. □

Figure 1 shows the law of a chain of nine spins below, at and above its crossing.

3.1 A Bessel bound at every N

The heuristic above replaced the number $\binom{a-1}{r}$ of ways to split a letters into $r+1$ runs by $a^r/r!$. After this change the run count is exactly the power series of I_0 and I_1. The next lemma bounds the error this makes.

Lemma 3.2 (clipped products)**.** *(a) If $t_1, \ldots, t_r \geq 0$ and $(v)_+ = \max(v, 0)$, then $0 \leq 1 - \prod_{k=1}^r (1 - t_k)_+ \leq \sum_{k=1}^r t_k$. (b) For integers $m \geq 1$ and $r \geq 0$, $U_r(m) := \frac{r!}{m^r}\binom{m-1}{r} = \prod_{k=1}^r \left(1 - \frac{k}{m}\right)_+$, so*

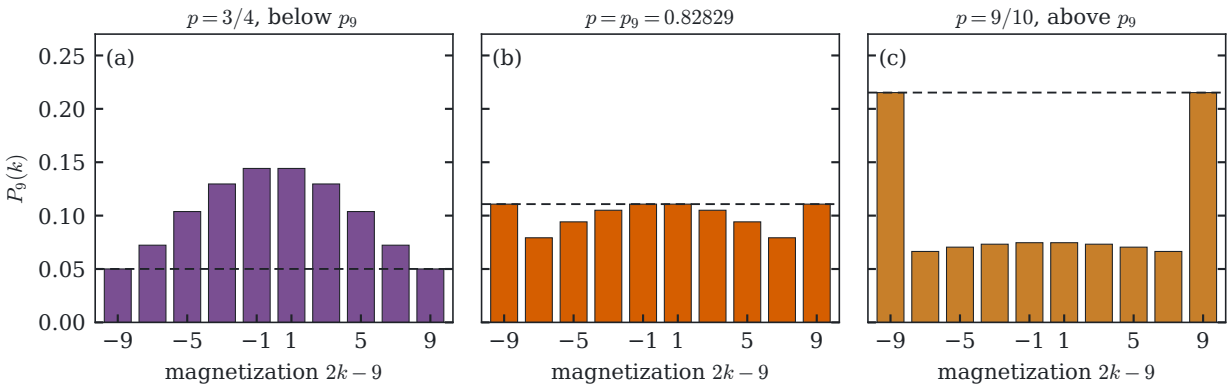


Figure 1: The bin probabilities $P_9(k)$ of a chain of nine spins against the magnetization $2k-9$, below (a), at (b) and above (c) the crossing $p_9 = 0.82829\ldots$, on a common vertical scale. The dashed line is the endpoint height. At $p = p_9$ the two central bins, with magnetization -1 and 1, are exactly as likely as each endpoint.

$0 \le 1 - U_r(m) \le \frac{r(r+1)}{2m}$. *(c) For* $x \ge 0$,

$$\sum_{r\ge 0} \frac{r(r+1)x^{2r}}{(r!)^2} = x^2 I_0(2x) + x I_1(2x),$$

$$\sum_{r\ge 0} \frac{r(r+1)x^{2r+1}}{r!(r+1)!} = x^2 I_1(2x), \qquad \sum_{r\ge 0} \frac{(r+1)x^{2r+1}}{r!(r+1)!} = x I_0(2x).$$

Proof. (a) Every factor lies in $[0,1]$, so the product is at most 1. If some $t_k \ge 1$ the product is 0 and the sum is at least 1. Otherwise $\prod_k (1-t_k) \ge 1 - \sum_k t_k$ by induction on r. (b) Note that $\frac{r!}{m^r}\binom{m-1}{r} = \prod_{k=1}^{r} \frac{m-k}{m}$. For $r < m$ every factor is positive, and for $r \ge m$ both sides vanish, the right side through the factor $k = m$. The bound is (a) with $t_k = k/m$. (c) Use the series of $I_0(2x)$ and $I_1(2x)$ from Section 1.2. Write $r(r+1) = r^2 + r$ in the first sum, cancel the factors r and $r+1$ against the factorials, and shift the index. □

Theorem 3.3 (uniform Bessel approximation)**.** *Put* $h = N/2$ *and* $F(x) = 2x\big(I_0(2x) + I_1(2x)\big)$, *and recall that* $u_N = (1-p_N)/p_N$.

(i) For every $N \ge 2$ *and* $x > 0$,

$$0 \le 1 - \frac{hS_N(x/h)}{F(x)} \le \frac{x(x+1)}{h}. \tag{7}$$

(ii) Let $x_N = hu_N$, *let* x_B *be the unique positive root of* $F(x_B) = h$, *and put* $\delta_N = x_N(x_N+1)/h$. *Then* $x_B \le x_N$, *and* $x_N - x_B \le -\frac{1}{2}\log(1-\delta_N)$ *whenever* $\delta_N < 1$.

(iii) With $u_B = x_B/h$ *we have* $0 \le u_N - u_B = O(L^2/N^2)$ *as* $N \to \infty$.

Section 3.2 uses part (i) to locate p_N at every N.

Proof. (i) *Step 1. The clipped factors.* Let $a = \lfloor N/2 \rfloor$ and $b = \lceil N/2 \rceil$. As in Lemma 3.2(b), $A_r = \frac{r!}{h^r}\binom{a-1}{r} = \prod_{k=1}^{r}(1 - \frac{h-a+k}{h})_+$, and B_r is the same with b in place of a. Since $a+b = 2h$ and $h-a \in \{0, \frac{1}{2}\}$, the $2r$ numbers $(h-a+k)/h$ and $(h-b+k)/h$ are nonnegative with sum $r(r+1)/h$, so Lemma 3.2(a) gives $0 \le 1 - A_r B_r \le r(r+1)/h$. In the same way the deficits of $A_{r+1}B_r$ and

A_rB_{r+1} are at most $((r+1)^2+h-a)/h$ and $((r+1)^2+h-b)/h$. Averaging cancels the parity offsets (the shifts $h-a$ and $h-b$ add to 0), so $0 \le 1-\frac{1}{2}(A_{r+1}B_r + A_rB_{r+1}) \le (r+1)^2/h$.

Step 2. Summing. With $u = x/h$ the run-count formula of Theorem 2.1 becomes

$$S_N(u) = 2u\Big[\sum_{r\ge 0}\frac{x^{2r}}{(r!)^2}A_rB_r + \sum_{r\ge 0}\frac{x^{2r+1}}{r!(r+1)!}\,\frac{A_{r+1}B_r + A_rB_{r+1}}{2}\Big].$$

Without the factors A_r, B_r the bracket is $I_0(2x)+I_1(2x)$ and the right side is $F(x)/h$. By Step 1 and Lemma 3.2(c), $F(x)/h - S_N(u)$ lies between 0 and $2u/h$ times $\sum_r r(r+1)\frac{x^{2r}}{(r!)^2} + \sum_r (r+1)^2\frac{x^{2r+1}}{r!(r+1)!} = x(x+1)\big(I_0(2x)+I_1(2x)\big)$. Dividing by $F(x)/h = 2u(I_0(2x)+I_1(2x))$ proves (i). This covers every r, including those beyond the degree of the polynomial, where the clipped products vanish and Lemma 3.2(a) still applies.

(ii) Differentiating the series term by term gives $F' = 2F + 2I_0(2x)$. So F increases from 0 to ∞, x_B is unique, and $(\log F)' \ge 2$. At $x = x_N$ we have $hS_N(u_N) = h$, so (i) gives $h \le F(x_N)$, hence $x_B \le x_N$, and $F(x_N) \le h/(1-\delta_N)$ when $\delta_N < 1$. Integrating $(\log F)' \ge 2$ from x_B to x_N gives $2(x_N - x_B) \le \log(F(x_N)/h) \le -\log(1-\delta_N)$.

Proof of Theorem 3.1(c). We use only part (i). Note that $x^{2r}/(r!)^2 \le (2x)^{2r}/(2r)!$ and $x^{2r+1}/(r!(r+1)!) \le (2x)^{2r+1}/(2r+1)!$ for $x \ge 0$, since $\binom{2r}{r} \le 4^r$ and $\binom{2r+1}{r} \le 2^{2r+1}$. So $I_0(2x)+I_1(2x) \le e^{2x}$ and $F(x) \le 2xe^{2x}$. Fix $\theta \in (0,1)$ and put $x_\pm = (1\pm\theta)L/2$. Then $F(x_-) \le (1-\theta)LN^{1-\theta} < h$ for large N, so $hS_N(x_-/h) < h$ by (i), and $x_- < x_N$ since S_N increases. On the other side $F(x_+) \ge 2x_+I_0(2x_+)$, which is at least a constant times $\sqrt{L}\,N^{1+\theta}$ for large N by the large-argument form of I_0, while $x_+(x_++1)/h \to 0$. So $hS_N(x_+/h) \ge F(x_+)(1-x_+(x_++1)/h) > h$ for large N, and $x_N < x_+$. Hence $Nu_N = 2x_N \sim L$, so $u_N \to 0$ and $z_N = Nu_N/(1+u_N) \sim L$, which is (c).

(iii) By Theorem 3.1(c), $x_N \sim L/2$. So $\delta_N = O(L^2/N)$, and (ii) gives $x_N - x_B = O(L^2/N)$. Divide by h. □

3.2 The crossing at every N

We measure the crossing by $z_N = N(1-p_N) = Nu_N/(1+u_N)$, the value of τ at $p = p_N$. Let $\psi(z) = z + \frac{1}{2}\log z$ and $c_0 = \frac{1}{2}\log(\pi/8) = -0.46735\ldots$, and let ζ_N be the positive root of

$$\psi(\zeta_N) = L + c_0, \quad \text{that is,} \quad 8\zeta_N e^{2\zeta_N} = \pi N^2 \quad \text{or} \quad \zeta_N = \tfrac{1}{2}W_0(\pi N^2/4), \tag{8}$$

with W_0 the principal branch of the Lambert function. This is the heuristic equation from the start of the section with w replaced by ζ. Since $\zeta e^{2\zeta}$ increases, ζ_N increases with N. Also $\zeta_N \le L$, since $\psi(L) \ge L + c_0$ when $L \ge \pi/8$. Finally let $\varepsilon_N = \zeta_N(\zeta_N+1)/N + 1/(16\zeta_N^2)$.

Theorem 3.4 (the crossing at every N)**.** *For every $N \ge 2$,*

$$\zeta_N + \frac{1}{8\zeta_N} - \varepsilon_N < z_N < \zeta_N + \frac{1}{8\zeta_N}.$$

The width ε_N tends to 0, since $\zeta_N \le L$ and $\zeta_N \to \infty$. It is 0.440, 0.163, 0.0387 and 0.00778 at $N = 10, 10^2, 10^3, 10^4$ (Table 1).

The two terms ζ_N and $1/(8\zeta_N)$ come from the following heuristic. Let $\Phi(w) = F(w/2) = w\big(I_0(w)+I_1(w)\big)$ and $\mathcal{E}(z) = \sqrt{\pi z/2}\,e^{-z}\big(I_0(z)+I_1(z)\big)$, which tends to 1 as $z \to \infty$. Put $w_* = Nu_N$ and $\Lambda_N = 2\Phi(w_*)/N$. At the crossing, (7) and (9) below give $\psi(w_*) + \log\mathcal{E}(w_*) = L + c_0 + \log\Lambda_N$ with $0 \le \log\Lambda_N \le -\log(1 - w_*(w_*+2)/(2N))$ whenever $w_*(w_*+2) < 2N$. With $\mathcal{E} = \Lambda_N = 1$ this

would say $w_* = \zeta_N$. Since $\log \mathcal{E}(w) = -1/(8w) + O(w^{-2})$ and $z_N = w_* + O(w_*^2/N)$, it suggests $z_N = \zeta_N + 1/(8\zeta_N) + O(\zeta_N^{-2} + \zeta_N^2/N)$. The theorem makes this error explicit and one-sided. Its proof uses two lemmas. Lemma 3.5 turns a comparison of z_N with a test value into a sign condition, and Lemma 3.6 bounds $\mathcal{E}$ with explicit constants.

Lemma 3.5 (root bracket)**.** *Let $N \geq 2$ and $w > 0$. Then*

$$\log \Phi(w) - \log \tfrac{N}{2} = \psi(w) - \psi(\zeta_N) + \log \mathcal{E}(w). \tag{9}$$

(a) If $\Phi(w) \leq N/2$, then $Nw/(N+w) \leq z_N$, strictly if $\Phi(w) < N/2$. (b) If $\Phi(w)\big(1-\frac{w(w+2)}{2N}\big) > N/2$, then $z_N < Nw/(N+w)$.

Proof. Note that $\log \Phi(w) = \psi(w) + \frac{1}{2}\log\frac{2}{\pi} + \log \mathcal{E}(w)$ and $\log\frac{N}{2} - \psi(\zeta_N) = -\log 2 - c_0 = \frac{1}{2}\log\frac{2}{\pi}$, which gives (9). Now apply (7) with $x = w/2$. Then $F(x) = \Phi(w)$ and $x(x+1)/h = w(w+2)/(2N)$, so

$$\Phi(w)\Big(1 - \frac{w(w+2)}{2N}\Big) \leq \frac{N}{2}\, S_N\Big(\frac{w}{N}\Big) \leq \Phi(w).$$

(a) The hypothesis gives $S_N(w/N) \leq 1 = S_N(u_N)$, strictly if $\Phi(w) < N/2$. Since S_N is strictly increasing, $w/N \leq u_N$, strictly in that case. The map $v \mapsto Nv/(1+v)$ increases and sends w/N to $Nw/(N+w)$ and u_N to z_N. (b) Now $S_N(w/N) > 1$, so $w/N > u_N$. □

Lemma 3.6 (Bessel bounds)**.** *For every $z > 0$,*

$$1 - \frac{1}{8z} - \frac{3}{128z^2} - \frac{45}{512z^3} \leq \mathcal{E}(z) \leq 1 - \frac{1}{8z} - \frac{3}{128z^2} + \frac{15}{512z^3},$$

and $\mathcal{E}(z) \leq 1$. In particular $\mathcal{E}(z) \leq 1 - \frac{1}{8z}$ for $z \geq \frac{5}{4}$.

The lemma is the large-argument expansion of $I_0 + I_1$ with explicit errors. We prove it in Appendix A. By [8, Eq. 10.32.3] and a substitution, $\mathcal{E}(z) = \pi^{-1/2}\int_0^{2z} e^{-v} v^{-1/2}\sqrt{1 - v/(2z)}\, dv$. We bound $\sqrt{1-y}$ above and below by cubics that share its Taylor terms up to y^2, extend the integral to ∞, and use $\Gamma(k+\frac{1}{2})/\Gamma(\frac{1}{2}) = 1, \frac{1}{2}, \frac{3}{4}, \frac{15}{8}$ for $k = 0, 1, 2, 3$.

Proof of Theorem 3.4. Step 1. The split at $N = 175$. By (8), $N = \sqrt{8\zeta_N/\pi}\, e^{\zeta_N}$, and the right side increases in ζ_N. Since $\sqrt{32/\pi}\, e^4 = 174.25\ldots$, we have $\zeta_N \geq 4$ exactly when $N \geq 175$. Write $\zeta = \zeta_N$ and $Y = \zeta + \frac{1}{8\zeta}$.

Step 2. The upper bound for $N \geq 175$. We apply Lemma 3.5(b) at $\bar{w} = NY/(N-Y)$, for which $N\bar{w}/(N+\bar{w}) = Y$. By (9) it suffices that $\mathcal{G} = \psi(\bar{w}) - \psi(\zeta) + \log \mathcal{E}(\bar{w}) + \log(1 - \bar{w}(\bar{w}+2)/(2N)) > 0$. Expanding each term with Lemma 3.6 gives $\mathcal{G} \geq \frac{1}{32\zeta^2} + \frac{\zeta(\zeta-1)}{2N} + O(\zeta^{-3} + N^{-1} + \zeta^4 N^{-2})$, and Appendix A makes every term explicit for $\zeta \geq 4$.

Step 3. The lower bound for $N \geq 175$. We apply Lemma 3.5(a) at $\bar{w}' = NY'/(N - Y')$ with $Y' = Y - \varepsilon_N$. It suffices that $\mathcal{G}' = \psi(\bar{w}') - \psi(\zeta) + \log \mathcal{E}(\bar{w}') < 0$, and we get $\mathcal{G}' \leq -\frac{3}{128\zeta^2} - \frac{\zeta}{N} + O(\zeta^{-3} + N^{-1} + \zeta^4 N^{-2})$, again with explicit constants for every $\zeta \geq 4$ in Appendix A. The term $-\zeta/N$ is why ε_N contains $\zeta(\zeta+1)/N$.

Step 4. Short chains, by certificates. Here $2 \leq N \leq 174$. Let $d = 10^9$, let $c = \lfloor du_N \rfloor$ and let $c' = c + 1$. At $N = 2$, where $du_2 = 5 \cdot 10^8$ is an integer, we take $c = du_2 - 1$ and $c' = c + 2$. The sign of $S_N(c/d) - 1$ is the sign of $n_N(m_N) - d^{N-1}$, where $n_N(m_N) = d^{N-1} S_N(c/d)$ is the integer of Remark 2.3 at $(\alpha, \beta) = (d, c)$ (Appendix A.3). We computed these integers at c and c' by two exact methods, the run-count sum of Theorem 2.1 and the recursion (3) with integer weights d, c in place of p, q. They agreed for every $N \leq 174$ and gave $c/d < u_N < c'/d$, so

$z(c/d) < z_N < z(c'/d)$ with $z(u) = Nu/(1+u)$. We then checked $\zeta_N + \frac{1}{8\zeta_N} - \varepsilon_N < z(c/d)$ and $z(c'/d) < \zeta_N + \frac{1}{8\zeta_N}$ with outward interval arithmetic at eighty decimal digits. To enclose π, we used $\pi = 16\arctan(1/5) - 4\arctan(1/239)$ and eighty terms of each alternating series, with the first omitted term bounding its remainder. Each arithmetic operation rounds outward. For a logarithm, we take the neighboring decimal on each side of the correctly rounded endpoint value.

Newton's method only proposes a value of ζ_N. We certify the bracket of half-width 10^{-45} around it by enclosing $\psi(\zeta) - L - c_0$ at both ends and checking opposite signs. Evaluating both comparisons on this whole bracket gives margins greater than 0.03319 (upper) and 0.08559 (lower), with the smallest at $N = 174$. The script `verify_certified_bounds.py` replays these checks from the stored rational endpoints, including all rounding and remainder bounds. Lean checks the short-chain comparisons independently with rational bounds on ζ_N and π. We also ran the certificates up to $N = 1000$ as a sanity check (both methods up to 400, then only the run-count sum), and they all pass. □

N	z_N	ζ_N	$z_N - \zeta_N - \frac{1}{8\zeta_N}$	$-\varepsilon_N$	R_N
2	0.666666666667	0.5368290974	−0.1030112	−0.62938	+0.050643
10	1.615958871376	1.6001732978	−0.0623310	−0.44048	−0.056660
100	3.503367368092	3.5100055489	−0.0422507	−0.16337	−0.043265
174	3.996779704730	3.9987131703	−0.0331935	−0.11878	−0.033450
175	4.001967989170	4.0038072816	−0.0330596	−0.11838	−0.033304
1000	5.590351166975	5.5807388611	−0.0127862	−0.03873	−0.011887
10^4	7.734265514999	7.7210118331	−0.0029359	−0.00778	−0.002051

Table 1: The crossing against Theorem 3.4, and the remainder R_N of (12). We computed z_N by run-count bisection in 50-digit decimal arithmetic and by a floating-point dynamic program, which agree to the digits shown.

The bracket is in terms of ζ_N (a value of the Lambert function). The next corollary states it for p_N and expands it in $L = \log N$.

Corollary 3.7. *Put $\nu_N = L - \frac{1}{2}\log L + c_0$, $Y_N = \zeta_N + \frac{1}{8\zeta_N}$ and $Y'_N = Y_N - \varepsilon_N$.*

(a) For every $N \geq 2$ we have $0 < Y'_N < Y_N < N$, $1 - Y_N/N < p_N < 1 - Y'_N/N$ and, for the coupling $J_N^{\mathrm{F}} = -\frac{1}{2}\log u_N$ at the crossing,

$$\frac{1}{2}\log\Big(\frac{N}{Y_N} - 1\Big) < J_N^{\mathrm{F}} < \frac{1}{2}\log\Big(\frac{N}{Y'_N} - 1\Big).$$

(b) For every $N \geq 2$, $\nu_N \leq \zeta_N \leq \nu_N + \frac{\log L - 2c_0}{4\nu_N}$ and

$$-\varepsilon_N < z_N - \nu_N < \frac{\log L - 2c_0 + \frac{1}{2}}{4\nu_N}. \tag{10}$$

In particular $z_N = L - \frac{1}{2}\log L + c_0 + O(\log L/L)$ as $N \to \infty$.

(c) As $N \to \infty$,

$$z_N = \nu_N + \frac{\frac{1}{4}\log L - \frac{1}{2}c_0 + \frac{1}{8}}{L} + O\Big(\frac{(\log L)^2}{L^2}\Big). \tag{11}$$

(d) (Computer-assisted.) $z_N > \zeta_N$ for every $N \geq 2774$.

Proof. (a) Since $\frac{1}{2} < \zeta_2 \le \zeta_N$, we have $\frac{1}{16\zeta_N^2} < \frac{1}{8\zeta_N}$, and since $\zeta_N \le L < N-1$, we have $\zeta_N(\zeta_N+1)/N < \zeta_N$. So $\varepsilon_N < Y_N$, that is, $Y'_N > 0$. Also $Y_N < L + \frac{1}{4} < N$. The bounds are Theorem 3.4 with $p_N = 1 - z_N/N$ and, by (2), $J_N^{\mathrm{F}} = \frac{1}{2}\log(N/z_N - 1)$, which decreases in z_N.

(b) By (8), $\zeta_N = L + c_0 - \frac{1}{2}\log\zeta_N$. Since $\zeta_N \le L$, this gives $\zeta_N \ge \nu_N$, and $\nu_N > 0$ since it increases in N and $\nu_2 > 0.40$. Also $\log y \le y - 1$ gives $\zeta_N - \nu_N = \frac{1}{2}\log(L/\zeta_N) \le (L - \zeta_N)/(2\zeta_N)$, and $L - \zeta_N = \frac{1}{2}\log\zeta_N - c_0 \le \frac{1}{2}\log L - c_0$. Hence $\zeta_N - \nu_N \le (\log L - 2c_0)/(4\nu_N)$. For z_N, add Theorem 3.4 and $0 < \frac{1}{8\zeta_N} \le \frac{1}{8\nu_N}$. The last claim follows since $\varepsilon_N = O(L^2/N + L^{-2})$.

(c) Put $\mu = \zeta_N - \nu_N = \frac{1}{2}\log(L/\zeta_N)$. By (b), $L - \zeta_N = \frac{1}{2}\log L - c_0 + O(\log L/L)$, so $\mu = \frac{L-\zeta_N}{2L} + O\big(\frac{(\log L)^2}{L^2}\big) = \frac{\frac{1}{4}\log L - \frac{1}{2}c_0}{L} + O\big(\frac{(\log L)^2}{L^2}\big)$. Also $\frac{1}{8\zeta_N} = \frac{1}{8L} + O(\frac{\log L}{L^2})$ and $\varepsilon_N = O(L^2/N + L^{-2})$, and Theorem 3.4 gives (c).

(d) By Theorem 3.4, $z_N - \zeta_N > \frac{1}{8\zeta_N} - \varepsilon_N$, which is positive when $f(\zeta_N) > 1$, where $f(\zeta) = n(\zeta)\big(\frac{1}{8\zeta} - \frac{1}{16\zeta^2}\big)/(\zeta(\zeta+1))$ and $n(\zeta) = \sqrt{8\zeta/\pi}\,e^{\zeta}$, so that $n(\zeta_N) = N$. The logarithmic derivative of f is $1 + \frac{1}{2\zeta} - \frac{2}{\zeta} + \frac{1}{2\zeta^2-\zeta} - \frac{1}{\zeta+1} > 1 - \frac{3}{2\zeta} - \frac{1}{\zeta+1} > 0$ for $\zeta \ge 3$. Since ζ_N increases with N, once $f(\zeta_N) > 1$ it stays above 1. This step is computer-assisted. With the outward intervals of Step 4, we obtain $f(\zeta_{2774}) - 1 > 2.0 \times 10^{-4}$ on the whole bracket. Here we evaluate $f(\zeta_N)$ by replacing $n(\zeta_N)$ with the exact integer N. Since $\zeta_{2774} > 3$, this proves (d). The same interval check gives $f(\zeta_{2773}) < 1$. The conclusion for every $N \ge 2774$ is also proved in Lean. That proof uses rational Taylor bounds for ζ_{2774} and an exponential comparison to extend the bound to larger N. □

The heuristic suggests that $\psi(z_N) \approx L + c_0 + 1/(8z_N)$. The next proposition proves this at every N with an explicit remainder.

Proposition 3.8 (the crossing equation)**.** *For every $N \ge 2$,*

$$z_N + \tfrac{1}{2}\log z_N = L + c_0 + \frac{1}{8z_N} + R_N, \qquad -\frac{z_N^2 + z_N}{N} \le R_N \le \frac{1}{16z_N^2}. \tag{12}$$

We also have $L/2 \le z_N \le L$, so $|R_N| \le \frac{1}{4L^2} + \frac{L^2+L}{N}$ for every $N \ge 2$.

We prove Proposition 3.8 in Appendix A. The proof writes R_N exactly, using (7) and (9) at $w = w_*$ (formula (41)). For $N \ge 175$, Lemma 3.6 puts the Bessel part of this formula between 0 and $0.0559/z_N^2$, and the other terms lie between $-(z_N^2 + z_N)/N$ and 0. For $N \le 174$ the proof uses the same certificates as Theorem 3.4.

Remark 3.9 (the sign of $z_N - \zeta_N$, computer-assisted). We have $z_N > \zeta_N$ for $N \le 15$, $z_N < \zeta_N$ for $16 \le N \le 218$, and $z_N > \zeta_N$ for every $N \ge 219$. For $N \le 2773$ this comes from the sign of $S_N - 1$ in exact integers at rationals on each side of $\zeta_N/(N - \zeta_N)$, the odds at which $z(u) = \zeta_N$, which the scripts list and replay with outward intervals (Section 6). The closest case is $N = 218$, where ζ_N is more than 1.65×10^{-5} from the certified bracket of z_N. For $N \ge 2774$ the sign is Corollary 3.7(d).

The ratios u_B/u_N of the Bessel root of Theorem 3.3 to the true odds are 0.9761, 0.9864, 0.9924 at $N = 100, 200, 400$ and 0.99579, 0.99769, 0.99874 at $N = 800, 1600, 3200$. So $1 - u_B/u_N$ is close to $L/(2N)$, which is 0.0230 at $N = 100$ and 0.00126 at $N = 3200$. This is in line with Theorem 3.3(iii), which gives a relative error $O(L/N)$ since $u_N \sim L/N$. Figure 2 compares the exact crossing with the expansions of Corollary 3.7 up to $N = 10^4$.

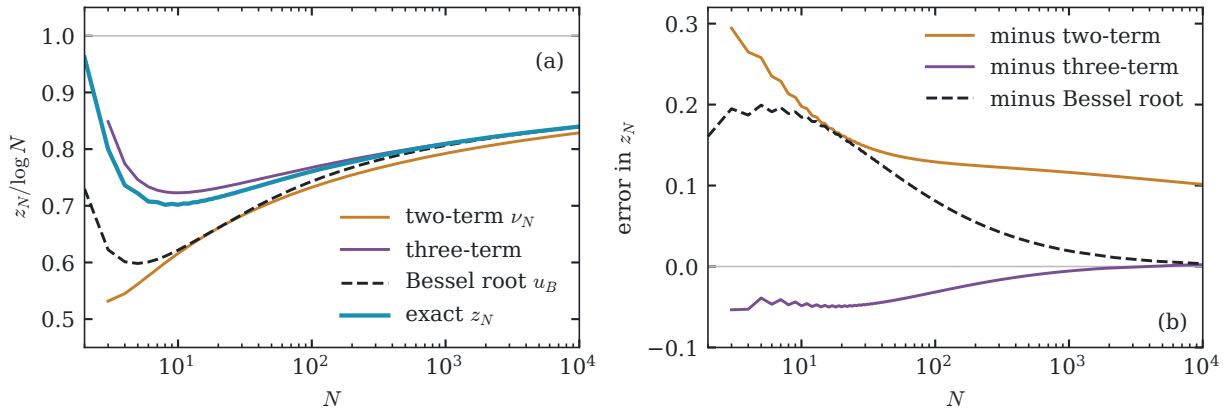


Figure 2: (a) The scaled crossing $z_N/\log N$ from the exact run polynomial (thick line), and the same quantity from the Bessel root u_B of Theorem 3.3 (dashed), from the two-term expansion ν_N of (10) and from the three-term expansion $\nu_N + (\frac{1}{4}\log L - \frac{1}{2}c_0 + \frac{1}{8})/L$ of (11). The ratio tends to 1 very slowly. (b) The exact z_N minus each approximation. The small oscillations for N below about 30 come from the parity of N.

4 The crossing of every bin

For each bin k we find the value of p at which bin k is exactly as likely as an endpoint. Figure 3 shows it for three values of N. The crossing $1 - p_{a,N-a}$ decreases as the bin moves in from the edge, from order $N^{-1/2}$ next to the edge to order $(\log N)/N$ in the middle.

4.1 Ordering the crossings

Let $a, b \geq 1$ be integers, $N = a + b$ and $p > 0$, and recall that $S_{a,b}(u) = P_N(a)/P_N(0)$ with $u = (1-p)/p$. Theorem 2.1 and $\binom{m-1}{r+1} = \frac{m-1-r}{r+1}\binom{m-1}{r}$ for $m = a, b$ give

$$S_{a,b}(u) = \sum_{r\geq 0} \binom{a-1}{r}\binom{b-1}{r}\Big(2u^{2r+1} + \frac{N-2-2r}{r+1}u^{2r+2}\Big). \tag{13}$$

Its coefficients are nonnegative, as $N - 2 - 2r \geq |b - a|$ for $r < \min(a, b)$, and the linear one is 2. So there is a unique $u_{a,b} > 0$ with $S_{a,b}(u_{a,b}) = 1$, and $p_{a,b} = 1/(1 + u_{a,b})$. As $S_{a,b}(u) \geq 2u$, we have $u_{a,b} \leq \frac{1}{2}$ and $p_{a,b} \in [\frac{2}{3}, 1)$, with $p_{a,b} = \frac{2}{3}$ only for $a = b = 1$.

Proposition 4.1 (ordering the crossings)**.** *If $a < b - 1$, then*

$$u_{a+1,b-1} < u_{a,b}, \qquad p_{a+1,b-1} > p_{a,b}.$$

Also $p_{a,b+1} > p_{a,b}$ for all positive a, b. In addition, $S_{a+1,b-1} \geq S_{a,b}$ and $S_{a,b+1} \geq S_{a,b}$ coefficientwise, with at least one strict coefficient.

So a bin closer to the middle, or in a longer row, needs a larger p to fall to the endpoint height.

Proof. For fixed N and $r \leq a-1$, $\binom{a-1}{r}\binom{b-1}{r} = (r!)^{-2}\prod_{j=0}^{r-1}(a-1-j)(b-1-j)$, a product of positive factors. Each factor grows when (a, b) becomes $(a+1, b-1)$, by $(a-j)(b-2-j)-(a-1-j)(b-1-j) = b - a - 1 > 0$. The even multiplier $(N - 2 - 2r)/(r + 1)$ is unchanged, and it is nonnegative where either product is nonzero, since then $r \leq a$ and $N - 2 - 2r \geq b - a - 2 \geq 0$. New summands are

nonnegative, and the cubic coefficient grows strictly, from $2(a-1)(b-1)$ to $2a(b-2)$. Increasing b with a fixed increases the binomial coefficients of Theorem 2.1, and the quadratic coefficient $N-2$ strictly. A polynomial with nonnegative coefficients that dominates another coefficientwise, strictly somewhere, is strictly larger at every $u>0$, so its root is smaller. □

The ordering of the bin heights in Proposition 4.2 also follows from the interior log-concavity of Dekking and Kong [12, Proposition 4.3], with the symmetry of the row, and they remark after that proposition that a sharper proof gives strict log-concavity. We use the coefficients because they give the strict inequalities directly and also compare different rows. The formula for p_N^{adj} in that proposition is already implicit in their work. By [12, Theorem 4.1], for $N \geq 4$ and $0<p<1$ the row is trimodal exactly when $P_N(0) > P_N(1)$, since $P_N(1) < P_N(2)$ here, and their Example 4.2 with $a=b=q$ and the stationary start gives $P_N(1)/P_N(0) = 2u+(N-2)u^2$, which is $S_{1,N-1}(u)$. The next proposition adds the comparison $p_N > p_N^{\mathrm{adj}}$ for $N \geq 4$ and the global extrema at p_N.

Proposition 4.2 (adjacent bins and global extrema)**.** *For $N \geq 2$ and $0<p<1$, an adjacent bin has the same probability as an endpoint precisely at*

$$p_N^{\mathrm{adj}} = \frac{1+\sqrt{N-1}}{2+\sqrt{N-1}}.$$

For $0<p<1$, the endpoints are the least likely bins if $p < p_N^{\mathrm{adj}}$, bins 1 and $N-1$ are if $p > p_N^{\mathrm{adj}}$, and they tie at equality. For $N \geq 4$, the central crossing satisfies $p_N > p_N^{\mathrm{adj}}$. At $p = p_N$, the middle bin and the endpoints attain the global maximum and bins $1, N-1$ attain the global minimum. For $N = 2, 3$, $p_N = p_N^{\mathrm{adj}}$ and every bin ties at the crossing.

So at the central crossing the row has the shape of a W. It is high at the two ends and in the middle, and lowest next to the ends.

Proof. By (13), $S_{1,N-1}(u) = 2u+(N-2)u^2$, with positive root $u^{\mathrm{adj}} = 1/(1+\sqrt{N-1})$. This gives p_N^{adj} and the sign change. For $0<p<1$ we have $P_N(k) = P_N(0)S_{k,N-k}(u)$ with $P_N(0)>0$, so if $1 \leq k < N-k-1$ Proposition 4.1 gives $P_N(k) < P_N(k+1)$. By symmetry the middle is the largest interior bin and bins $1, N-1$ are the smallest, which gives the claims about the least likely bins. For $N \geq 4$ the cubic coefficient $2(\lfloor N/2 \rfloor - 1)(\lceil N/2 \rceil - 1)$ of S_N is positive, so $2u_N + (N-2)u_N^2 < S_N(u_N) = 1$, hence $u_N < u^{\mathrm{adj}}$ and $p_N > p_N^{\mathrm{adj}}$. At p_N the middle ties with the endpoints, which gives the global maximum, and bins $1, N-1$ are below the endpoints, which gives the global minimum. For $N = 2, 3$ we have $S_N = S_{1,N-1}$, and symmetry gives the ties. □

So $1 - p_N^{\mathrm{adj}} \sim N^{-1/2}$, much larger than $1 - p_N \sim (\log N)/N$. Away from the ends, the single letter + of bin 1 needs two switches (one into it and one out of it), so $S_{1,N-1} \approx Nu^2$ reaches 1 only at $u \approx N^{-1/2}$. (At $p=0$ and $p=1$ there are also ties between bins of probability 0.)

4.2 A Bessel estimate uniform in the bin

Words in bin a with $r+1$ runs of each letter have $2r+1$ switches. Their contribution to $S_{a,b}$ is about $2u(abu^2)^r/(r!)^2$. This explains the variable abu^2, so a bin near the edge acts like the middle of a shorter row with $\sqrt{ab}$ in place of $N/2$. Put

$$s = \sqrt{ab}, \quad A = \frac{ab}{a+b}, \quad c = \frac{a+b}{2\sqrt{ab}}, \quad \eta = \frac{1}{c}, \quad x = su, \quad \mathcal{J}_c(x) = I_0(2x) + cI_1(2x),$$

and $\widehat{S}_{a,b}(u) = 2u\mathcal{J}_c(su)$. Here $c \geq 1$ and $0 < \eta \leq 1$, with $\eta = 1$ exactly when $a = b$. This Bessel form is classical. The fixed-scale law of Herrmann and Vallois [5, Remark 3.4] with a fair start has atoms $e^{-\tau}/2$ at ± 1 and density $\frac{1}{2}\tau e^{-\tau}\{I_0(\tau\rho) + I_1(\tau\rho)/\rho\}$ at magnetization per spin m, where $\rho = \sqrt{1-m^2}$. At $m = (a-b)/N$ we have $\rho = 2s/N$, and with τ replaced by Nu, the density times the bin width $2/N$, divided by an atom, is $\widehat{S}_{a,b}(u)$. The next theorem bounds how far the exact ratio is from it, at every N.

Theorem 4.3 (uniform estimate for every bin)**.** *For all positive integers a, b and all $u > 0$,*

$$0 \leq 1 - \frac{S_{a,b}(u)}{\widehat{S}_{a,b}(u)} \leq \frac{(a+b)u^2}{2} + u = \frac{x^2}{2A} + \frac{x}{s}. \tag{14}$$

In particular, for fixed N and u the bound does not depend on the bin. The inequality still holds when the right side exceeds 1, but then it gives no lower bound on $S_{a,b}$.

When u is about L/N, the bound is $O(L^2/N)$ for every bin at once. The proof follows that of Theorem 3.3(i), with the two letters weighted by a and b.

Proof. Let $U_r = U_r(a)$ and $V_r = U_r(b)$ as in Lemma 3.2(b), and note that $1/a + 1/b = 1/A$. Lemma 3.2(a), applied to the $2r$ clipped factors of $U_r V_r$, gives

$$0 \leq 1 - U_r V_r \leq \frac{r(r+1)}{2A}. \tag{15}$$

Lemma 3.2(a) gives $1 - U_{r+1}V_r \leq \frac{(r+1)(r+2)}{2a} + \frac{r(r+1)}{2b}$, and the same with a and b exchanged for $U_r V_{r+1}$. Averaging with weights a and b, the weighted average $E_r = (aU_{r+1}V_r + bU_r V_{r+1})/(a+b)$ satisfies

$$\begin{aligned} 0 \leq 1 - E_r &\leq \frac{(r+1)(r+2)}{a+b} + \frac{r(r+1)}{2(a+b)}\Big(\frac{a}{b} + \frac{b}{a}\Big) \\ &= \frac{r(r+1)}{2A} + \frac{2(r+1)}{a+b}. \end{aligned} \tag{16}$$

Since $\binom{a-1}{r} = a^r U_r / r!$, (13) becomes

$$S_{a,b}(u) = 2u\Big[\sum_{r\geq 0} \frac{x^{2r}}{(r!)^2} U_r V_r + c\sum_{r\geq 0} \frac{x^{2r+1}}{r!(r+1)!} E_r\Big], \tag{17}$$

and dropping $U_r V_r$ and E_r gives $\widehat{S}_{a,b}(u)$. So by Lemma 3.2(c), with $I_j = I_j(2x)$, $(\widehat{S}_{a,b} - S_{a,b})/(2u)$ lies between 0 and

$$\frac{1}{2A}(x^2 I_0 + x I_1 + c x^2 I_1) + \frac{2c\,x I_0}{a+b} = \Big(\frac{x^2}{2A} + \frac{x}{s}\Big)\mathcal{J}_c(x),$$

since $c/s = 1/(2A)$ and $2c/(a+b) = 1/s$. Divide by $\widehat{S}_{a,b}/(2u) = \mathcal{J}_c(x)$. □

To turn this into a bound on the crossing, let $x_{a,b} = s u_{a,b}$ and let $\hat{x}_{a,b}$ be the positive root of

$$G_\eta(\hat{x}_{a,b}) = A, \qquad G_\eta(x) = x\{I_1(2x) + \eta I_0(2x)\}. \tag{18}$$

Differentiating term by term gives

$$G'_\eta(x) - 2G_\eta(x) = 2x(1-\eta)\{I_0(2x) - I_1(2x)\} + \eta I_0(2x) \geq 0, \tag{19}$$

since $I_0(2x) - I_1(2x) = \frac{1}{\pi}\int_0^\pi e^{2x\cos\theta}(1 - \cos\theta)\,d\theta > 0$ by [8, Eq. 10.32.3]. So $(\log G_\eta)' \geq 2$ for $x > 0$ and $0 \leq \eta \leq 1$, and $\hat{x}_{a,b}$ is unique.

Corollary 4.4 (quantitative crossing bound)**.** *We have $x_{a,b} \geq \hat{x}_{a,b}$. If*

$$\delta_{a,b} = \frac{x_{a,b}^2}{2A} + \frac{x_{a,b}}{s} < 1,$$

then

$$0 \leq x_{a,b} - \hat{x}_{a,b} \leq -\tfrac{1}{2}\log(1-\delta_{a,b}). \tag{20}$$

Proof. Since $c\eta = 1$ and $2c/s = 1/A$, we have $\widehat{S}_{a,b}(u) = G_\eta(su)/A$. At the crossing $S_{a,b} = 1$, so Theorem 4.3 gives $A \leq G_\eta(x_{a,b}) \leq A/(1-\delta_{a,b})$. As $G_\eta(\hat{x}_{a,b}) = A$, this gives $x_{a,b} \geq \hat{x}_{a,b}$, and integrating $(\log G_\eta)' \geq 2$ from $\hat{x}_{a,b}$ to $x_{a,b}$ gives (20). This is the argument of Theorem 3.3(ii). For even N and $a = b$ the two statements coincide, since then $\eta = 1$, $s = h$ and $2G_1 = F$, so $\hat{x}_{a,b} = x_B$. □

Remark 4.5 (Jacobi polynomials). The run sums of this paper can be written in terms of Jacobi polynomials, and their Bessel and Laguerre forms are limits of Jacobi polynomials near the point 1. For $a \leq b$ and $0 \leq t < 1$,

$$\sum_{r\geq 0}\binom{a-1}{r}\binom{b-1}{r}t^r = {}_2F_1(1-a,1-b;1;t) = (1-t)^{a-1}P_{a-1}^{(0,b-a)}\Big(\frac{1+t}{1-t}\Big),$$

and for $b > a$ and $0 \leq u < 1$,

$$\sum_{r\geq 0}\binom{a-1}{r}\binom{b-1}{r+1}u^{2r+2} = \frac{(b-1)u^2}{a}(1-u^2)^{a-1}P_{a-1}^{(1,b-a-1)}\Big(\frac{1+u^2}{1-u^2}\Big).$$

For fixed $\alpha, \beta > -1$, the Mehler–Heine limit [8, Eq. 18.11.5] gives $n^{-\alpha}P_n^{(\alpha,\beta)}(1+w^2/(2n^2)) \to (w/2)^{-\alpha}I_\alpha(w)$ as $n \to \infty$, uniformly for bounded $w \geq 0$, with the value at $w = 0$ taken by continuity. Elliott [19, Section 3, Eq. (3.10)] gives formal modified-Bessel expansions near 1 from the exterior, for fixed parameters with $\alpha \geq 0$. For fixed a and $y \geq 0$, put $u^2 = y/b$ and $\beta = b-a-1$. The argument of the second Jacobi polynomial is $1 - 2x_b/\beta$, where $x_b = -\beta y/(b-y) \to -y$ as $b \to \infty$. The fixed-degree Jacobi–Laguerre limit [8, Eq. 18.7.21] is locally uniform in its argument, by coefficient convergence of polynomials of fixed degree. It makes the last sum tend to $(y/a)L_{a-1}^{(1)}(-y) = H_a(y)$ of (25). The odd-run contribution, including its factor $2u$, and the other even-run contribution are respectively $O_a(b^{-1/2})$ and $O_a(b^{-1})$, uniformly for y in compact sets. Temme [20, Section 4.2] also compares Jacobi with Laguerre polynomials, and Gil, Segura and Temme [21, Section 2 and Remark 1] expand $P_n^{(\alpha,\beta)}(1-2x/\beta)$ in inverse powers of β with Laguerre coefficients, uniformly for bounded n, α and x (complex x included). This covers the fixed degree $a-1$ of Section 4.4, while Theorem 4.7 lets the degree grow, uniformly in $b \geq a$. Also, Theorem 4.8 is about the root of $H_a(y) = 1$, a level of the Laguerre polynomial at the negative argument $-y$, and not about its zeros. Elliott defers truncation-error bounds to a proposed later paper [19, Section 1]. Here $b-a$ can be as large as $N-2$, and Theorem 4.3 gives an explicit relative error at every N.

4.3 Every growing distance from an edge

The Bessel root satisfies an equation of the form $z + \frac{1}{2}\log z = \ell + C + B/z + \dots$. The next lemma inverts such equations.

Lemma 4.6 (inverting $z+\frac{1}{2}\log z$)**.** *Fix $K>0$. There is a constant C_K with the following property. Let $z\ge 1$ and $\ell\ge 2$, and suppose that $z+\frac{1}{2}\log z=\ell+C+B/z+\omega$ with $|C|,|B|\le K$ and $|\omega|\le 1$. Then*

$$\Big|z-\ell+\tfrac{1}{2}\log\ell-C-\frac{\frac{1}{4}\log\ell-\frac{1}{2}C+B}{\ell}\Big|\le C_K\Big(\frac{(\log\ell)^2}{\ell^2}+|\omega|\Big).$$

Proof. Put $w=z-\ell$. As $z\ge 1$, $\log z\ge 0$ and $|B/z|\le K$, so $z\le \ell+2K+1$ and $|w|\le 2K+1+\frac{1}{2}\log(\ell+2K+1)=O_K(\log\ell)$. This proves the claim for ℓ bounded in terms of K, since then both sides are $O_K(1)$ and $(\log\ell)^2/\ell^2$ is bounded below. For larger ℓ we have $|w|\le \ell/2$, so $\log z=\log\ell+w/\ell+O((w/\ell)^2)$ and $B/z=B/\ell+O_K(|w|/\ell^2)$, and $w=C-\frac{1}{2}\log\ell-w/(2\ell)+B/\ell+\omega+O_K((\log\ell)^2/\ell^2)$. This gives first $w=d+O_K(\log\ell/\ell+|\omega|)$ with $d=C-\frac{1}{2}\log\ell$, and then, putting this back into the term $w/(2\ell)$, $w=d+(B-d/2)/\ell+O_K((\log\ell)^2/\ell^2+|\omega|)$. □

Theorem 4.7 (uniform crossing expansion)**.** *Let $1\le a\le b$, and put*

$$\ell=\log A,\qquad C(\eta)=\tfrac{1}{2}\log(8\pi)-\log(1+\eta),\qquad B(\eta)=\frac{3-\eta}{8(1+\eta)}.$$

As $a\to\infty$, uniformly over $b\ge a$,

$$\begin{aligned}2\sqrt{ab}\,u_{a,b}&=\ell-\tfrac{1}{2}\log\ell+C(\eta)\\&\quad+\frac{\frac{1}{4}\log\ell-\frac{1}{2}C(\eta)+B(\eta)}{\ell}+O\left(\frac{(\log\ell)^2}{\ell^2}+\frac{(\log a)^2}{a}\right).\end{aligned}\tag{21}$$

That is, there are a_0 and K' such that the error term is at most K' times the bracket for all $a\ge a_0$ and all $b\ge a$. The same holds with $1-p_{a,b}$ in place of $u_{a,b}$.

Here $A=ab/(a+b)$ is half the harmonic mean of a and b. There is no condition on how a compares with N, so the theorem covers a bin at distance $\sqrt{N}$ from the edge as well as the middle. The proof uses Corollary 4.4 to pass to the Bessel root $\hat{x}_{a,b}$. The large-argument form of I_0 and I_1 then turns (18) into an equation that Lemma 4.6 solves.

Proof. Here $a/2\le A\le a$, $s\ge a$ and $0<\eta\le 1$.

Step 1. The crossing is at $x_{a,b}\le\log a$. At $x=\log a$ the bound of Theorem 4.3 is at most $2(\log a)^2/a$, while $\widehat{S}_{a,b}=G_\eta(x)/A\ge xI_1(2x)/a\to\infty$, both uniformly in b (the identity $\widehat{S}_{a,b}(u)=G_\eta(su)/A$ is in the proof of Corollary 4.4). So $S_{a,b}>1$ there, and since $S_{a,b}$ increases, $x_{a,b}\le\log a$ for all $a\ge a_0$ and all $b\ge a$. Then $\delta_{a,b}\le 2(\log a)^2/a$, and Corollary 4.4 gives

$$x_{a,b}-\hat{x}_{a,b}=O((\log a)^2/a).\tag{22}$$

Step 2. The Bessel root. Write $z=2\hat{x}_{a,b}$, so $z\to\infty$ as $A\to\infty$, uniformly in η. By the large-argument form of I_0 and I_1 [8, Eq. 10.40.1], $I_1(z)+\eta I_0(z)=(1+\eta)e^z(2\pi z)^{-1/2}(1-B(\eta)/z+O(z^{-2}))$ uniformly in $0\le\eta\le 1$. Taking logarithms in (18) gives $z+\frac{1}{2}\log z=\ell+C(\eta)+B(\eta)/z+O(z^{-2})$. As $|C(\eta)|\le 2$ and $0<B(\eta)\le\frac{3}{8}$, Lemma 4.6 with $K=2$ expands z.

Step 3. Back to the crossing. By (22) the expansion moves to $2x_{a,b}=2\sqrt{ab}\,u_{a,b}$. Finally $2s(u-u/(1+u))\le 2x^2/s=O((\log a)^2/a)$ at $u=u_{a,b}$, which gives the same expansion for $1-p_{a,b}=u_{a,b}/(1+u_{a,b})$. □

Hence, uniformly in b,

$$1-p_{a,b}\sim\frac{\log a}{2\sqrt{ab}}\qquad(a\to\infty,\ b\ge a).\tag{23}$$

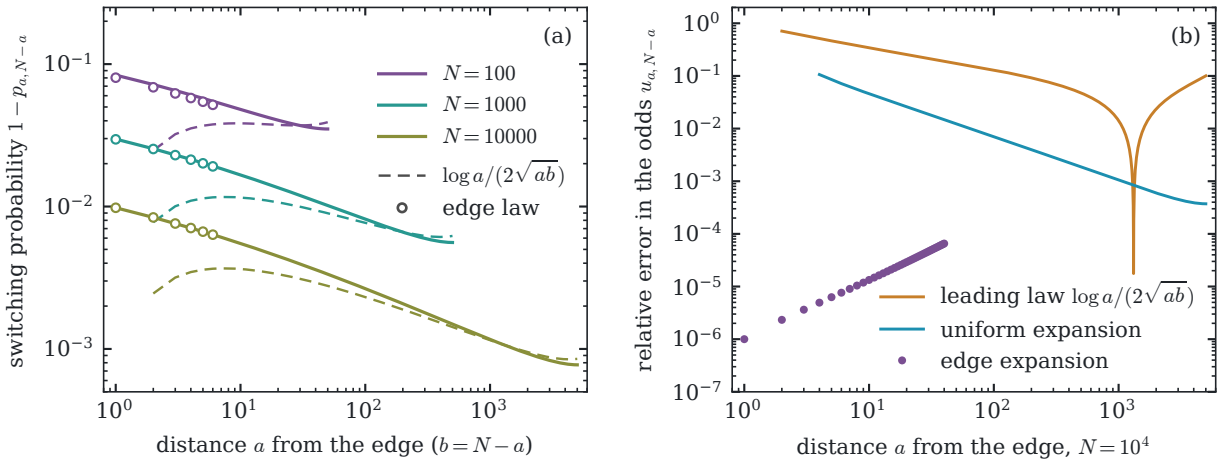


Figure 3: (a) The switching probability $1-p_{a,N-a}$ at the crossing of each bin, from the exact run polynomial, against its distance a from the nearer edge, for $N = 100$, 1000 and 10^4. The dashed lines are the leading law (23). It holds as a grows, so it turns away from the exact curves near the edge. The circles are the edge law (27) for $a \le 6$. The right end of each line is the central crossing $1-p_N$. (b) For $N = 10^4$, the relative error in the odds $u_{a,N-a}$ of the leading law $\log a/(2\sqrt{ab})$, of the main terms of the uniform expansion (21) for $a \ge 4$, and of the first three terms of the edge expansion (26) for $a \le 40$. The error of the leading law changes sign near $a = 1300$.

If $a = o(b)$, then $\eta \to 0$ and $\log A = \log a + o(1)$, so

$$2\sqrt{ab}(1-p_{a,b}) = \log a - \tfrac{1}{2}\log\log a + \tfrac{1}{2}\log(8\pi) + o(1). \tag{24}$$

If $a/N \to \alpha \in (0, \frac{1}{2}]$ at any rate, then

$$2\sqrt{a(N-a)}(1-p_{a,N-a}) = \log N - \tfrac{1}{2}\log\log N + \log\{\alpha(1-\alpha)\} + C(2\sqrt{\alpha(1-\alpha)}) + o(1).$$

At the center, $a = b = N/2$ gives $A = N/4$ and $\eta = 1$, and changing from $\log A$ to $\log N$ recovers (11).

4.4 Fixed bins and the Laguerre edge law

In Ising terms, bin a with a fixed is magnetization $-N + 2a$, a few up spins in a sea of down spins, and we ask when it is as likely as full order. Now let a be fixed and $b \to \infty$. A run of letters $+$ away from the ends needs two switches, which gives a factor u^2, and it can go in about b places. So the right variable is $y = bu^2$. Define

$$H_a(y) = \sum_{r=0}^{a-1} \binom{a-1}{r} \frac{y^{r+1}}{(r+1)!} = \frac{y}{a} L_{a-1}^{(1)}(-y), \tag{25}$$

with the standard Laguerre polynomial $L_{a-1}^{(1)}$. Let y_a be the positive root of $H_a(y_a) = 1$ and $c_a = \sqrt{y_a}$. Then $y_1 = 1$, $y_2 = \sqrt{3} - 1$, and y_a decreases strictly in a, by coefficient comparison.

Theorem 4.8 (fixed-edge expansion)**.** *For each fixed positive integer a, as $b \to \infty$,*

$$u_{a,b} = \frac{c_a}{\sqrt{b}} - \frac{1}{b} + \frac{\gamma_a}{b^{3/2}} + O_a(b^{-2}), \tag{26}$$

where

$$\gamma_a = \frac{1 + y_a + (y_a + y_a^2/2)H_a''(y_a)/H_a'(y_a)}{2c_a}.$$

Thus the second coefficient in the odds expansion is -1 *for every fixed bin. For the switching probability,*

$$1 - p_{a,b} = \frac{c_a}{\sqrt{b}} - \frac{1 + c_a^2}{b} + O_a(b^{-3/2}). \tag{27}$$

For bin 1, where $y_1 = 1$ and $H_1'' = 0$, this gives $u_{1,b} = b^{-1/2} - b^{-1} + b^{-3/2} + \dots$, the expansion of the exact root $1/(1 + \sqrt{b})$ of Proposition 4.2.

Proof. Put $\theta = b^{-1/2}$, $y = bu^2$ and $V_r = U_r(b)$. For $b \geq a$, (13) reads

$$\begin{aligned} S_{a,b}(u) = 2\sqrt{y/b} \sum_r \binom{a-1}{r} \frac{y^r}{r!} V_r \\ + \frac{y}{b} \sum_r \binom{a-1}{r+1} \frac{y^r}{r!} V_r + y \sum_r \binom{a-1}{r} \frac{y^r}{(r+1)!} V_{r+1}. \end{aligned} \tag{28}$$

These sums have at most a terms, and with $V_r = 1$ they are $H_a'(y)$, $H_a''(y)$ and $H_a(y)$. Expanding $V_r = 1 - r(r+1)/(2b) + O_a(b^{-2})$ and using $\sum_r (r+1)(r+2)\binom{a-1}{r}\frac{y^{r+1}}{(r+1)!} = y^2 H_a''(y) + 2yH_a'(y)$ gives, uniformly near $y = y_a$,

$$\begin{aligned} S_{a,b}(u) = H_a(y) + 2\sqrt{y}\, H_a'(y)\theta \\ + \{(y - y^2/2)H_a''(y) - yH_a'(y)\}\theta^2 + O_a(\theta^3). \end{aligned} \tag{29}$$

With $b = \theta^{-2}$ the right side of (28) is analytic in (θ, y) near $(0, y_a)$, and $H_a'(y_a) > 0$. So the implicit-function theorem gives an analytic solution $y(\theta) = y_a + d_1\theta + d_2\theta^2 + O_a(\theta^3)$ of $S_{a,b} = 1$, and by the uniqueness of the crossing it is the value of y at $u = u_{a,b}$. Matching coefficients in (29) gives $d_1 = -2c_a$ and $d_2 = 2 + y_a + (y_a + y_a^2/2)H_a''(y_a)/H_a'(y_a)$. Expanding $u = \theta\sqrt{y}$ and then $u/(1+u)$ proves both expansions. □

The next theorem shows that for large a the Laguerre root y_a obeys the growing-distance law (24).

Theorem 4.9 (matching the edge and Bessel regimes)**.** *For every positive integer* a *and every* $x > 0$*,*

$$0 \leq 1 - \frac{aH_a(x^2/a)}{xI_1(2x)} \leq \frac{x^2}{2a}. \tag{30}$$

Consequently, with $\ell = \log a$ *and* $C(0) = \frac{1}{2}\log(8\pi)$ *as in Theorem 4.7, as* $a \to \infty$*,*

$$\begin{aligned} 2\sqrt{ay_a} = \ell - \tfrac{1}{2}\log\ell + C(0) \\ + \frac{\frac{1}{4}\log\ell - \frac{1}{2}C(0) + 3/8}{\ell} + O\left(\frac{(\log\ell)^2}{\ell^2} + \frac{(\log a)^2}{a}\right). \end{aligned} \tag{31}$$

Proof. With $U_r = U_r(a)$, $aH_a(x^2/a) = x\sum_{r\geq 0} U_r x^{2r+1}/(r!(r+1)!)$, so Lemma 3.2(b) and the second identity of Lemma 3.2(c) give (30). Put $x_a = \sqrt{ay_a}$, so $aH_a(x_a^2/a) = a$, and $x_a < \log a$ for large a by (30). So $a \leq G_0(x_a) \leq a/(1 - x_a^2/(2a))$, and as in Corollary 4.4, $x_a - x_0 = O((\log a)^2/a)$, where $G_0(x_0) = a$. So $z = 2x_0$ satisfies $z + \frac{1}{2}\log z = \ell + C(0) + 3/(8z) + O(z^{-2})$ by [8, Eq. 10.40.1], as in the proof of Theorem 4.7 with $\eta = 0$ and $A = a$, and Lemma 4.6 gives the expansion. □

Figure 3 compares these laws with the exact crossings.

4.5 The moving boundary layer

The crossings also give the shape of a whole row at odds $u = v_N > 0$. Between the central crossing (v_N about $(\log N)/N$) and the adjacent one (v_N about $N^{-1/2}$), the middle is above the endpoints and a layer of bins next to each edge is below them. (This is a layer of bins near each end of the row, that is, in the magnetization, not a region of the chain.) The width of this layer is

$$\mathcal{L}_N = \#\{1 \le k \le \lfloor N/2 \rfloor : P_N(k) < P_N(0)\}.$$

Corollary 4.10 (boundary-layer width)**.** *Suppose that, as $N \to \infty$,*

$$\Omega_N = \frac{1}{\sqrt{N}\, v_N} \longrightarrow \infty, \qquad \frac{\log \Omega_N}{N v_N} \longrightarrow 0.$$

Then

$$\mathcal{L}_N \sim \Omega_N^2 \log^2 \Omega_N = \frac{\log^2(1/(\sqrt{N}\, v_N))}{N v_N^2}. \tag{32}$$

In particular, if $v_N = N^{-\kappa}$ with $1/2 < \kappa < 1$, then

$$\mathcal{L}_N \sim (\kappa - \tfrac{1}{2})^2 N^{2\kappa - 1} \log^2 N.$$

By symmetry, the total number of interior bins below the endpoint height is $2\mathcal{L}_N$ for all sufficiently large N.

Proof. Set $d_N = \Omega_N^2 \log^2 \Omega_N$. Then $d_N \to \infty$ and $d_N/N = (\log \Omega_N/(N v_N))^2 \to 0$. Fix $\theta \in (0,1)$ and integers $a_\pm = (1 \pm \theta) d_N + O(1)$. Then $\log a_\pm \sim 2 \log \Omega_N$ and $\sqrt{a_\pm (N - a_\pm)}\, v_N \sim \sqrt{1 \pm \theta}\, \log \Omega_N$, so (23) gives $u_{a_\pm, N - a_\pm}/v_N \to 1/\sqrt{1 \pm \theta}$. Bin k is below the endpoint exactly when $v_N < u_{k,N-k}$, and $u_{k,N-k}$ decreases strictly for $k \le N/2$ by Proposition 4.1. So $a_- \le \mathcal{L}_N < a_+$ for large N, and we let $\theta \downarrow 0$. Finally $\mathcal{L}_N = o(N)$, so the reflected intervals are disjoint. □

Figure 4 shows one row inside this regime and the width $\mathcal{L}_N$ for three exponents κ.

5 Periodic boundaries and the crossing window

We now close the sequence of steps into a loop, so that the last step also interacts with the first. With this closing bond the spins form the periodic Ising chain, with weight proportional to $\exp(J \sum_{i=1}^{N} \sigma_i \sigma_{i+1})$, $\sigma_{N+1} = \sigma_1$ and again $u = e^{-2J}$. Antal, Droz and Rácz [9] give its magnetization law at a fixed scale, and Dantchev and Rudnick [13] give exact coefficients. We find how much the closing bond moves the central crossing, and then describe the row near the crossing under both boundaries.

Write $S^{\mathrm{F}}_{N,k} = S_{k,N-k}$ and $S^{\mathrm{P}}_{N,k}$ for the bin-to-endpoint ratios under free and periodic boundaries. Put $a = k$, $b = N - k$ and $s = \sqrt{ab}$. Counting cyclic words by their runs gives

$$S^{\mathrm{P}}_{N,k}(u) = \sum_{r=1}^{\min(a,b)} \frac{N}{r} \binom{a-1}{r-1} \binom{b-1}{r-1} u^{2r}. \tag{33}$$

To see this, note that a cyclic word with a letters $+$ and b letters $-$ switches an even number $2r$ of times and has r runs of each letter. Choose compositions of a and of b into r positive runs and a position for the start of a distinguished run of letters $+$, and divide by the r choices of that run.

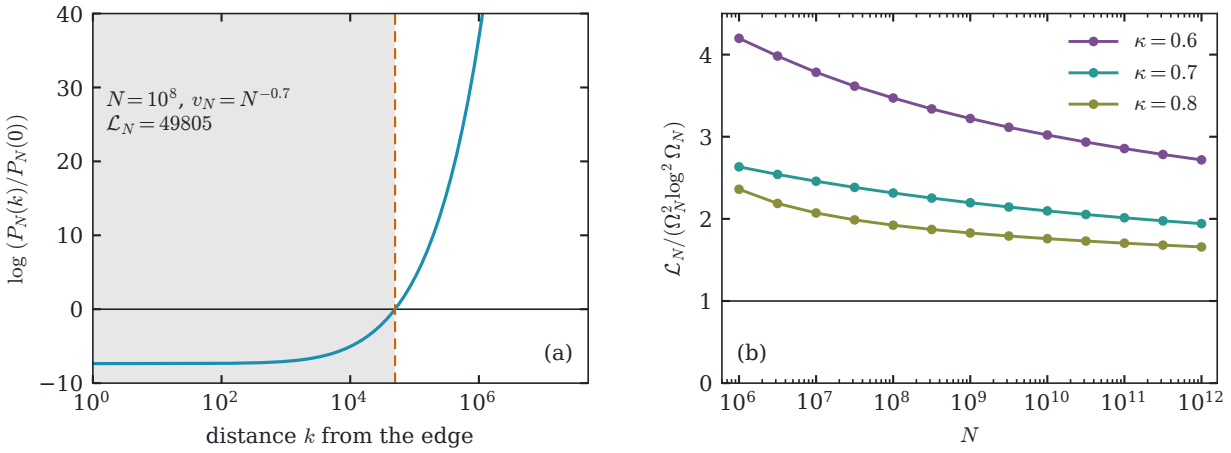


Figure 4: (a) The ratio $\log(P_N(k)/P_N(0))$ of each bin to an endpoint, from the exact run polynomial, against the distance k from the nearer edge, at $N = 10^8$ and odds $v_N = N^{-0.7}$. The shaded bins are below the endpoint height, and there are $\mathcal{L}_N = 49805$ of them in the left half. (b) The width $\mathcal{L}_N$ divided by its leading term $\Omega_N^2 \log^2 \Omega_N$ of (32), for $v_N = N^{-\kappa}$ with $\kappa = 0.6$, 0.7 and 0.8. The ratios decrease toward 1 only slowly. If $\mathcal{L}_N$ is instead divided by the distance a at which the main terms of (21) equal v_N, the ratio is within 0.013 of 1 for every N shown.

The normalization of the periodic measure cancels in the ratio. In particular $S^{\mathrm{P}}_{N,1} = Nu^2$, which gives the adjacent crossing $J = (\log N)/4$ of Stepanyan et al. [1]. Every interior ratio increases strictly from 0 to ∞ in $u > 0$, and the ratios increase strictly toward the middle, apart from the symmetric middle tie, by the coefficient comparison of Proposition 4.1.

Theorem 5.1 (periodic relative error)**.** *For $N \geq 2$, $1 \leq k \leq N-1$, and $u > 0$, define*

$$\widehat{S}^{\mathrm{P}}_{N,k}(u) = \frac{Nu}{\sqrt{k(N-k)}} I_1\Bigl(2u\sqrt{k(N-k)}\Bigr).$$

Then

$$0 \leq 1 - \frac{S^{\mathrm{P}}_{N,k}(u)}{\widehat{S}^{\mathrm{P}}_{N,k}(u)} \leq \frac{Nu^2}{2}. \tag{34}$$

So under periodic boundaries the ratio of every bin to an endpoint is close to a single Bessel function I_1, with the same kind of error as in Theorem 4.3.

Proof. Put $x = su$, $A = ab/N$, $U_r = U_r(a)$ and $V_r = U_r(b)$. Shifting the index in (33) gives $S^{\mathrm{P}}_{N,k}(u) = (Nu/s)\sum_{r\geq 0} U_r V_r x^{2r+1}/(r!(r+1)!)$, so the ratio to $\widehat{S}^{\mathrm{P}}_{N,k}$ is an average of the $U_r V_r$ with weights $x^{2r+1}/(r!(r+1)!)$, whose sum is $I_1(2x)$. By (15), $0 \leq 1 - U_r V_r \leq r(r+1)/(2A)$, and the second identity of Lemma 3.2(c) bounds the deficit by $x^2/(2A) = Nu^2/2$. □

We first give the idea behind the next theorem. A cyclic word switches an even number of times (it must come back to its first letter), so only I_1 appears in the periodic approximation, while the free words that start and end on different letters give the extra I_0 term of the free one. As $I_0 \approx I_1$ for large arguments, the free ratio is about twice the periodic one near the crossing. Both grow like e^{Nu}, so the periodic crossing needs Nu about $\log 2$ larger. Antal, Droz and Rácz [9, Eqs. (37) and (47)] work at a fixed value of $\tau = Nq$ (their scaling variable $\zeta = Nt/2$ with $t = 1 - \tanh J$,

which equals τ). There the endpoint atoms are $1/(2\cosh\tau)$ (periodic) and $e^{-\tau}/2$ (free), which again differ by a factor of about 2 for large τ, while their densities at zero magnetization agree to leading order as $\tau \to \infty$.

Theorem 5.2 (periodic crossing and boundary shift)**.** *Let u_N^{P} be the positive root of $S_{N,m_N}^{\mathrm{P}}(u) = 1$, let $p_N^{\mathrm{P}} = 1/(1+u_N^{\mathrm{P}})$, and let $u_N^{\mathrm{F}} = u_N$ be the free central root. Put $c_{\mathrm{P}} = \frac{1}{2}\log(\pi/2)$. As $N \to \infty$,*

$$Nu_N^{\mathrm{P}} = L - \tfrac{1}{2}\log L + c_{\mathrm{P}} + \frac{\frac{1}{4}\log L - \frac{1}{2}c_{\mathrm{P}} + \frac{3}{8}}{L} + O\!\left(\frac{(\log L)^2}{L^2}\right), \tag{35}$$

$$N(u_N^{\mathrm{P}} - u_N^{\mathrm{F}}) = \log 2 + \frac{\frac{1}{4} - \frac{1}{2}\log 2}{\log N} + o(1/\log N). \tag{36}$$

Replacing Nu_N^{P} by $N(1-p_N^{\mathrm{P}})$ preserves (35). At the periodic crossing the coupling $J_N^{\mathrm{P}} = -\frac{1}{2}\log u_N^{\mathrm{P}}$ satisfies

$$J_N^{\mathrm{P}} = \tfrac{1}{2}(\log N - \log\log N) + \frac{\frac{1}{4}\log\log N - \frac{1}{2}c_{\mathrm{P}}}{\log N} + O\!\left(\frac{(\log\log N)^2}{(\log N)^2}\right). \tag{37}$$

Since $c_{\mathrm{P}} = c_0 + \log 2$, the periodic expansion is the free one shifted by $\log 2$, with $\frac{3}{8}$ in place of $\frac{1}{8}$ in the $1/L$ term.

Proof. Step 1. The first order. At the middle $2s = N + O(1/N)$. At $u = (1 \pm \theta)L/N$ with fixed $\theta \in (0,1)$ the bound of Theorem 5.1 tends to 0, and the large-argument form of I_1 makes $\widehat{S}^{\mathrm{P}}$ tend to ∞ with the plus sign and to 0 with the minus sign. Since S_{N,m_N}^{P} increases in u, we get $Nu_N^{\mathrm{P}} \sim L$.

Step 2. The crossing equation. Write $z = Nu_N^{\mathrm{P}}$. At the root $S^{\mathrm{P}} = 1$, so $\log \widehat{S}^{\mathrm{P}} = O(L^2/N)$ by (34). Replacing the argument $2su$ by z, and the factor Nu/s by $2z/N$, changes $\log \widehat{S}^{\mathrm{P}}$ by $O(L/N^2)$. Here $(\log I_1)' = I_0/I_1 - 1/z = O(1)$ for large positive arguments by [8, Eqs. 10.29.2, 10.40.1]. Also $I_1(z) = e^z(2\pi z)^{-1/2}(1 - 3/(8z) + O(z^{-2}))$ [8, Eq. 10.40.1]. So

$$z + \tfrac{1}{2}\log z = L + c_{\mathrm{P}} + \frac{3}{8z} + O(z^{-2} + L^2/N),$$

and Lemma 4.6 with $\ell = L$, $C = c_{\mathrm{P}}$ and $B = \frac{3}{8}$ gives (35). As $N(u - u/(1+u)) = O(L^2/N)$ for $Nu = O(L)$, it also holds for $N(1 - p_N^{\mathrm{P}})$.

Step 3. The shift and the coupling. By (2), $Nu_N = z_N + O(L^2/N)$. Subtracting (11), where $c_0 = c_{\mathrm{P}} - \log 2$ and the correction is $\frac{1}{8}$, gives (36), even with error $O((\log L)^2/L^2)$. Finally $J = -\frac{1}{2}\log u = \frac{1}{2}(L - \log z)$, and $\log z = \log L - (\frac{1}{2}\log L - c_{\mathrm{P}})/L + O((\log L)^2/L^2)$ gives (37). □

Before we return to the coupling (37) at the end of this section, we describe the whole row near the crossing. The factor e^{-2y^2} in the next theorem has a simple heuristic. Moving $yN/\sqrt{\log N}$ away from the middle multiplies the Bessel argument by $2\sqrt{k(N-k)}/N \approx 1 - 2y^2/\log N$. The argument is about $\log N$, so it drops by about $2y^2$, and the ratio, which grows like e to the argument, drops by about the factor e^{-2y^2}.

Theorem 5.3 (spatial crossing window)**.** *Let $X \in \{\mathrm{F}, \mathrm{P}\}$ and let u_N^X be its exact central crossing. For fixed real t, y set*

$$u = u_N^X + \frac{t}{N}, \qquad k_N(y) = \operatorname{nint}\!\left(\frac{N}{2} + \frac{yN}{\sqrt{\log N}}\right),$$

where either nearest-integer convention may be used. Then, as $N \to \infty$, uniformly on compact sets of (t, y),

$$S^X_{N,k_N(y)}(u^X_N + t/N) \longrightarrow e^{t-2y^2}. \tag{38}$$

If $M^X_N(t)$ is the number of interior bins with probability strictly above the endpoint probability at that parameter, then

$$\frac{\sqrt{\log N}}{N} M^X_N(t) \longrightarrow \sqrt{2\max(t, 0)}. \tag{39}$$

So the crossing window has width about $1/N$ in u and about $N/\sqrt{\log N}$ in space, and inside it the ratio of each bin to an endpoint is a Gaussian bump times e^t. A weak limit theorem (a limit for the overall shape of the law) cannot give this, since it says nothing about a single bin such as the endpoint. The Gaussian itself is expected. At $m = 2y/\sqrt{\log N}$ the factor e^{-2y^2} is $e^{-m^2 \log N/2}$, the large-τ Gaussian $e^{-m^2\tau/2}$ of Antal, Droz and Rácz [9, Section IV.E, Eq. (51)] at $\tau \approx \log N$. Theorem 5.3 adds the comparison with the endpoint, centered at the exact crossing and uniform on compact sets.

Proof. Step 1. The profile. Put $\lambda = k/N - \frac{1}{2}$ and $\eta = 2\sqrt{k(N-k)}/N = \sqrt{1 - 4\lambda^2}$. For the stated rounding and bounded y,

$$\eta = 1 - \frac{2y^2}{L} + O\Big(L^{-2} + \frac{1}{N\sqrt{L}}\Big), \qquad N u^X_N = L + O(\log L).$$

So the Bessel argument $Nu\eta$ at (u, k) is $Nu^X_N + t - 2y^2 + o(1)$, uniformly on the compact set. By Theorems 4.3 and 5.1 both ratios are within relative error $O(L^2/N)$ of their Bessel forms, and $I_0(z)$ and $I_1(z)$ equal $e^z(2\pi z)^{-1/2}(1 + O(1/z))$ [8, Eq. 10.40.1], so the prefactors relative to their central values tend to 1 uniformly. The change of argument gives e^{t-2y^2}, and the exact central ratio is 1, which proves (38).

Step 2. The count. The interior ratios increase toward the middle, by Proposition 4.1 for $X = \mathrm{F}$ and by the coefficient comparison after (33) for $X = \mathrm{P}$. If $t \le 0$, the central ratio is at most 1, so no interior bin is above the endpoint. Let $t > 0$ and $0 < \theta < \sqrt{t/2}$. By Step 1, for large N the bins at scaled distance less than $\sqrt{t/2} - \theta$ are above the endpoint and those at distance $\sqrt{t/2} + \theta$ are below, and by monotonicity so are all farther bins. Count the bins in between, divide by $N/\sqrt{L}$ and let $\theta \downarrow 0$. Rounding changes the count by $O(1)$. □

Remark 5.4 (constrained free energies). Let $Z^X_N(M)$ be the sum of the Ising weights of the configurations with magnetization M under boundary X, and $F^X_N(M) = -\log Z^X_N(M)$ its dimensionless free energy, as in the fixed-magnetization ensemble of [13, 15]. The normalization cancels in the ratio of a bin to an endpoint, so $-\log S^X_{N,k} = F^X_N(2k - N) - F^X_N(N)$. On a compact set of (t, y) the limit e^{t-2y^2} in (38) is bounded below by a positive constant, so the convergence there also holds for logarithms, uniformly. Hence

$$F^X_N(2k_N(y) - N) - F^X_N(N) \longrightarrow 2y^2 - t$$

at the odds $u^X_N + t/N$. In words, near the crossing the free energy of a fixed magnetization, measured from full magnetization, is a parabola in the scaled magnetization, lowered by t. This compares single bins. Each of them has probability of order $\sqrt{\log N}/N$ there, so it says nothing about the total weight of the nearly ordered configurations.

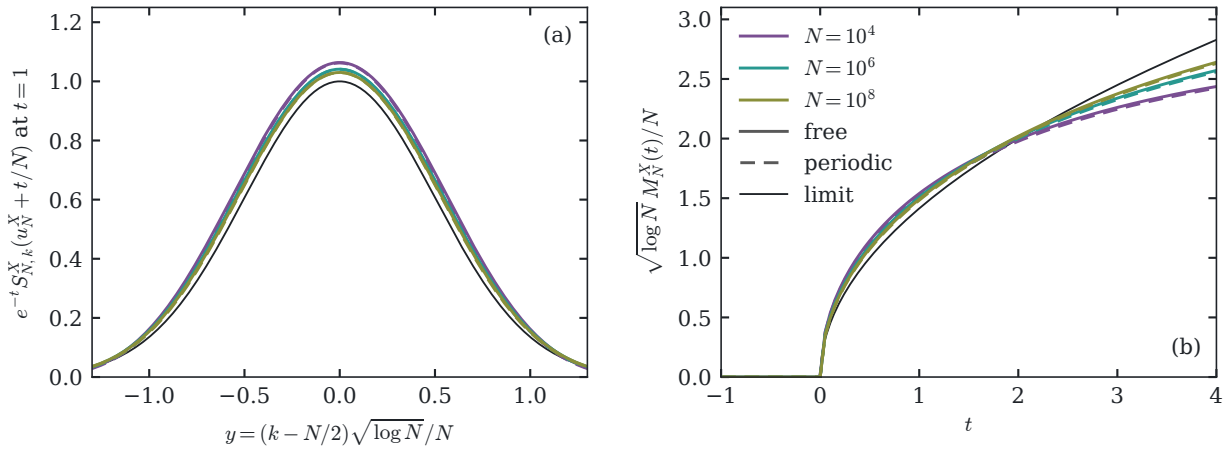


Figure 5: (a) The profile $e^{-t}S^X_{N,k}(u^X_N + t/N)$ at $t = 1$ against $y = (k - N/2)\sqrt{\log N}/N$, for free (solid) and periodic (dashed) boundaries and $N = 10^4$, 10^6 and 10^8, with the limit e^{-2y^2} of Theorem 5.3 (thin black line). (b) The scaled number $\sqrt{\log N}\, M^X_N(t)/N$ of interior bins above the endpoint height, with the limit $\sqrt{2\max(t, 0)}$. All curves come from the exact run polynomials at the exact central crossings. The convergence is slow. At $t = 2$ and $y = 0$ the relative error of the profile is 0.12 at $N = 10^4$, 0.08 at $N = 10^6$ and 0.06 at $N = 10^8$, under either boundary.

Figure 5 shows the exact ratios and counts for N from 10^4 to 10^8.

We end with the coupling at the two central crossings, since this is how Stepanyan et al. [1] state the crossing. Write $J^{\mathrm{F}}_N = -\frac{1}{2}\log u_N$ for the coupling at the free central crossing. For $J > 0$ let $\xi(J) = -1/\log\tanh J$ be the correlation length of the infinite chain, so that $\langle\sigma_0\sigma_r\rangle = e^{-r/\xi(J)}$ there. This is the zero-field correlation length with no constraint on the magnetization, evaluated at the coupling of the crossing. The spin correlations in a chain of fixed magnetization behave differently, and can even become negative [22, Sections II and IV].

Corollary 5.5 (couplings and correlation lengths at the crossings)**.** *As $N \to \infty$,*

(a) The free coupling satisfies $J^{\mathrm{F}}_N = \frac{1}{2}(\log N - \log\log N) + \dfrac{\frac{1}{4}\log\log N - \frac{1}{2}c_0}{\log N} + O\Big(\dfrac{(\log\log N)^2}{(\log N)^2}\Big)$.

(b) The two couplings differ by $J^{\mathrm{F}}_N - J^{\mathrm{P}}_N = \dfrac{\log 2}{2\log N} + O\Big(\dfrac{\log\log N}{(\log N)^2}\Big)$.

(c) The correlation lengths satisfy $N/\xi(J^{\mathrm{F}}_N) = 2\log N - \log\log N + \log(\pi/8) + O(\log\log N/\log N)$, *and* $N/\xi(J^{\mathrm{P}}_N) = 2\log N - \log\log N + \log(\pi/2) + O(\log\log N/\log N)$.

All three parts are facts about Nu at the crossings, read on other scales. The coupling is $-\frac{1}{2}\log u$, so it sees Nu only through $\log(Nu)$. The shift of $\log 2$ in Nu between the two boundary conditions therefore becomes a shift of order $1/\log N$ in the coupling. The correlation length is $1/(2u)$ to leading order, so N/ξ is about $2Nu$.

Proof. (a) By (2), $u_N = z_N/(N - z_N)$, so $J^{\mathrm{F}}_N = \frac{1}{2}(L - \log z_N) + O(L/N)$. Taking logarithms in (11) gives $\log z_N = \log L - (\frac{1}{2}\log L - c_0)/L + O((\log L)^2/L^2)$, and (a) follows in the same way as (37) in Step 3 of the proof of Theorem 5.2.

(b) The difference is $J^{\mathrm{F}}_N - J^{\mathrm{P}}_N = \frac{1}{2}\log(u^{\mathrm{P}}_N/u^{\mathrm{F}}_N) = \frac{1}{2}\log(1 + \delta)$, where $\delta = N(u^{\mathrm{P}}_N - u^{\mathrm{F}}_N)/(Nu^{\mathrm{F}}_N)$. By (36) the numerator is $\log 2 + O(1/L)$. By (2) and Corollary 3.7(b) the denominator is $z_N + O(L^2/N) = L + O(\log L)$. So $\delta = (\log 2)/L + O(\log L/L^2)$, and $\frac{1}{2}\log(1 + \delta) = \delta/2 + O(\delta^2)$.

(c) With $u = e^{-2J}$ we have $\tanh J = (1-u)/(1+u)$, so $N/\xi(J) = N \log \frac{1+u}{1-u} = 2Nu + O(Nu^3)$ for $0 < u \le \frac{1}{2}$. At either crossing $Nu^3 = O(L^3/N^2)$. It remains to insert $Nu_N = z_N + O(L^2/N) = L - \frac{1}{2}\log L + c_0 + O(\log L/L)$, from Corollary 3.7(b), and the expansion (35) of Nu_N^{P}, and to recall that $2c_0 = \log(\pi/8)$ and $2c_{\mathrm{P}} = \log(\pi/2)$. □

So at either crossing the correlation length is about $N/(2\log N)$. The chain is ordered over long stretches, but each stretch is a vanishing fraction of the chain, and N/ξ grows like $2\log N$. Antal, Droz and Rácz [9] give the magnetization law when τ is fixed, and there N/ξ stays bounded (it is about 2τ). The crossing lies outside that regime, which is why we need bounds that are uniform in τ. Part (b) is the shift by $\log 2$ in Nu of (36), seen on the logarithmic scale of the coupling. So the boundary condition moves the coupling at the crossing by an amount that decays only like $1/\log N$.

Stepanyan et al. [1, Section III.B] call the periodic central crossing the second thermal transition of the finite chain. They give the approximate formula $\beta J \approx (2\log N - 1)/5$ [1, Eq. (16)], and their Fig. 1(b) shows the fitted line $\beta J \approx 0.3864 \log N - 0.21055$. Table 2 and Figure 6 compare both with the exact crossing and with (37) without its error term. The line is accurate for $10 \le N \le 100$, but neither approximation has a $\log\log N$ term. The main terms of (37) have slope $\frac{1}{2} - 1/(2\log N)$ in $\log N$, up to $O(\log\log N/(\log N)^2)$, so the exact curve bends slowly toward slope $\frac{1}{2}$. Its secant slope is 0.385 on $10 \le N \le 10^2$, close to the fitted 0.3864, then 0.417 on $10^2 \le N \le 10^3$, and 0.450 on $10^4 \le N \le 10^5$. At the tabulated sizes $N = 10^3, 10^4, 10^5, 10^6$, the main terms of (37) are closer to the exact value than either approximation, and at $N = 10^5$ they are off by 0.002, against 0.171 and 0.338.

Remark 5.6 (other comparisons). We compare one central bin with one endpoint, as Stepanyan et al. [1] do. Comparing instead with both endpoints, or more generally solving $S^X_{N,m_N}(u) = r$ for a fixed $r > 0$, only adds $\log r$ to the constant. For periodic ends, Step 2 of the proof of Theorem 5.2 goes through with $\log r$ added to the right side of its crossing equation. For free ends the same argument works with Theorem 3.3(i) and (9) in place of Theorem 5.1, and with $\frac{1}{8}$ in place of $\frac{3}{8}$. So (35) and (11) hold for these roots with $c_X + \log r$ in place of c_X, and the coupling moves by $-(\log r)/(2\log N)$ to leading order. For $r = 2$ the free constant becomes $c_0 + \log 2 = c_{\mathrm{P}}$, so this $\log 2$, which comes from the convention, should not be confused with the shift by $\log 2$ between the boundary conditions in Theorem 5.2. For odd N, adding the two central bins as well as the two endpoints gives $r = 1$ again. The bracket of Theorem 3.4 is proved only for $r = 1$.

We end with what the crossing looks like in terms of domain walls. Given that the free chain is in bin m_N, a word has probability proportional to $u^{\mathrm{chg}(w)}$, so the number of domain walls has the law of the coefficients of S_N. Let $\mathcal{W}_N(u) = uS_N'(u)/S_N(u)$ be its mean, and put $\chi(w) = I_0(w)/(I_0(w) + I_1(w))$.

Proposition 5.7 (domain walls at the crossing). *Let $N \ge 2$ and $u > 0$, put $w = Nu$, and suppose that $w(w+2) < 2N$. Let $\lambda = -\log(1 - w(w+2)/(2N))$. Then for every $\delta > 0$,*

$$\bigl|\mathcal{W}_N(u) - w - \chi(w)\bigr| \le w(e^{\delta} - 1) + \frac{\lambda}{\delta}.$$

At the crossing, $\mathcal{W}_N(u_N) = z_N + \frac{1}{2} + \frac{1}{8z_N} + O\bigl((\log N)^{-2}\bigr)$.

In words, the central bin at the crossing carries one half more domain wall than the switching scale. Numerically, $\mathcal{W}_N(u_N) - z_N$ is 0.5191, 0.5166 and 0.5132 at $N = 10^3, 10^4, 10^5$, against 0.5224, 0.5162 and 0.5126 for $\frac{1}{2} + \frac{1}{8z_N}$.

Proof. Put $s = \log u$, $x = hu = w/2$ with $h = N/2$, $g(s) = \log S_N(e^s)$ and $G(s) = \log F(he^s)$. Both are logarithms of power series in e^s with nonnegative coefficients, so both are convex in s, and $g'(s) = \mathcal{W}_N(u)$. By Theorem 3.3(i), $G(s) - \log h - \lambda \le g(s) \le G(s) - \log h$, and the upper bound holds at every s. From $F' = 2F + 2I_0(2x)$ we get $G'(s) = xF'(x)/F(x) = w + \chi(w)$. The function χ decreases, since I_1/I_0 is the mean of $\cos t$ under the density proportional to $e^{w\cos t}$ on $[0,\pi]$ [8, Eq. 10.32.3], and its derivative in w is the variance of $\cos t$, which is positive. By convexity,

$$\mathcal{W}_N(u) \le \frac{g(s+\delta) - g(s)}{\delta} \le \frac{G(s+\delta) - G(s) + \lambda}{\delta} \le G'(s+\delta) + \frac{\lambda}{\delta} \le we^{\delta} + \chi(w) + \frac{\lambda}{\delta},$$

and in the same way $\mathcal{W}_N(u) \ge we^{-\delta} + \chi(w) - \lambda/\delta$. Since $1 - e^{-\delta} \le e^{\delta} - 1$, this proves the bound.

At the crossing $w = Nu_N = z_N + O(L^2/N)$ and $\lambda = O(L^2/N)$, and $\delta = (L/N)^{1/2}$ makes the bound $O(L^{3/2}N^{-1/2})$. The large-argument forms of I_0 and I_1 give $\chi(w) = \frac{1}{2} + \frac{1}{8w} + O(w^{-2})$, which proves the second claim. □

Since $u = e^{-2J}$, the definition of $\mathcal{W}_N$ gives

$$\frac{d}{dJ} \log \frac{P_N(m_N)}{P_N(0)} = -2\mathcal{W}_N(u).$$

The energy $-\sum_{i<N} \sigma_i\sigma_{i+1}$ of a configuration exceeds that of the all-up configuration by twice its number of domain walls. So at the crossing the mean energy gap between the central bin and the all-up configuration is $2\mathcal{W}_N(u_N) \approx 2\log N$, and over a range of couplings of width $1/(2\mathcal{W}_N(u_N))$, which is 0.061 at $N = 10^4$ and 0.048 at $N = 10^5$, the ratio of the central bin to an endpoint changes by about a factor e. The same proof with Theorem 5.1 in place of Theorem 3.3, where G becomes $\log(xI_1(2x))$ and $G' = wI_0(w)/I_1(w)$, gives the mean number of domain walls $Nu + \frac{1}{2} + O(1/(Nu))$ at the periodic crossing for even N.

Remark 5.8 (full order against one bin). The chain is fully ordered, all up or all down, with probability $P_N(0) + P_N(N) = p^{N-1}$. This is $\frac{1}{2}$ at $u_{\mathrm{ord}} = 2^{1/(N-1)} - 1$, that is, at the coupling $J_{\mathrm{ord}} = \frac{1}{2}\log\frac{N-1}{\log 2} + O(1/N)$. By Corollary 3.7(a), $J_N^{\mathrm{F}} = \frac{1}{2}\log(N/z_N - 1)$, so

$$J_{\mathrm{ord}} - J_N^{\mathrm{F}} = \tfrac{1}{2}\log\frac{z_N}{\log 2} + O\Big(\frac{\log N}{N}\Big) = \tfrac{1}{2}\log\log N - \tfrac{1}{2}\log\log 2 + o(1).$$

The difference is 1.046 at $N = 10^3$ and 1.330 at $N = 10^5$. So full order wins against the rest of the chain only at a coupling about $\frac{1}{2}\log\log N$ above the central crossing. The $\log\log N$ in J_N^{F} measures how much less likely one central bin is than the whole disordered band.

N	J_N^{P}	(37)	[1, (16)]	[1, Fig. 1(b)]	J_N^{F}	$J_N^{\mathrm{F}} - J_N^{\mathrm{P}}$
10	0.687	0.776	0.721	0.679	0.823	0.136
10^2	1.574	1.597	1.642	1.569	1.658	0.084
10^3	2.534	2.541	2.563	2.459	2.591	0.057
10^4	3.540	3.543	3.484	3.348	3.582	0.042
10^5	4.576	4.578	4.405	4.238	4.610	0.033
10^6	5.633	5.634	5.326	5.128	5.661	0.028

Table 2: The couplings J_N^{P} and J_N^{F} at the periodic and free central crossings, from the exact run polynomials and again from the transfer matrix (the two agree to 10^{-12}), against the main terms of (37) and the two approximations of Stepanyan et al. [1] (their Eq. (16) and the line of their Fig. 1(b)). By Corollary 5.5(b) the last column is $(\log 2)/(2 \log N)$ to leading order, which is 0.030 at $N = 10^5$. Differences are computed before rounding.

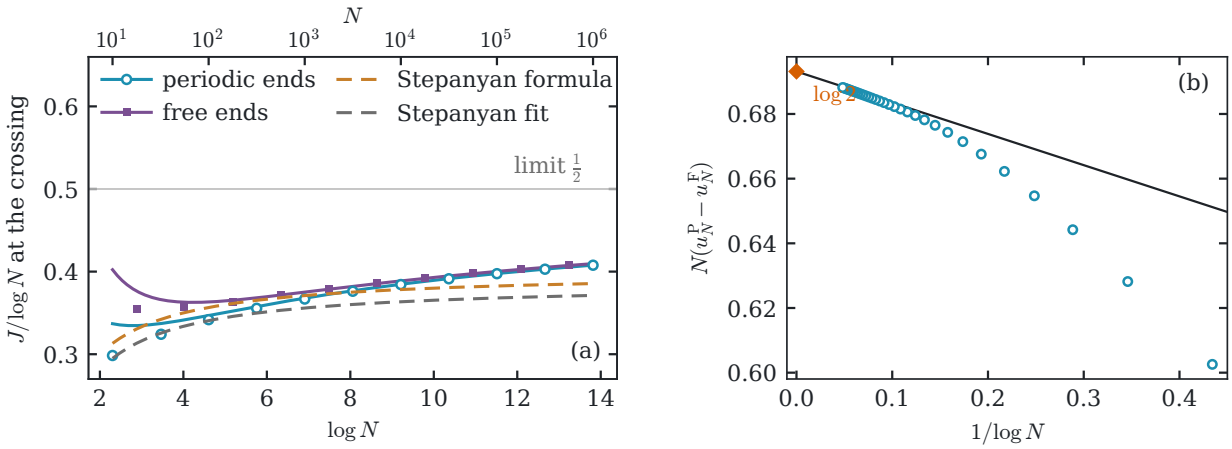


Figure 6: (a) The coupling at the central crossing divided by $\log N$, against $\log N$, from the exact run polynomials for periodic (circles) and free (squares) ends, with the main terms of (37) and of Corollary 5.5(a) (solid lines of the same color) and the two approximations of Stepanyan et al. [1] (dashed). The thin line is the limit $\frac{1}{2}$. The two approximations have no $\log \log N$ term and a slope in $\log N$ below $\frac{1}{2}$, so they drift away. (b) The boundary shift $N(u_N^{\mathrm{P}} - u_N^{\mathrm{F}})$ for $10 \le N \le 10^9$, from the exact run polynomials, against $1/\log N$, with the first two terms $\log 2 + (\frac{1}{4} - \frac{1}{2}\log 2)/\log N$ of (36) (line). The shift tends to $\log 2$ (diamond), and at $N = 10^9$ it differs from the two terms by 3×10^{-4}.

6 Verification

Aside from Remark 3.9, which is computer-assisted throughout, the proofs use two computations, namely the integer certificates for $N \le 174$ in Theorem 3.4 (Step 4) and Proposition 3.8, and the evaluation at $N = 2774$ in Corollary 3.7(d). Each one is described where it is used. The scripts are Python, and all but one use only the standard library. They check the exact law, the orderings, the Bessel bounds, the asymptotic formulas and the couplings of Table 2, and where possible each reported number is computed in two independent ways. For instance, the couplings of Table 2 come from the run polynomials and again from the transfer matrix of the chain (with NumPy), and the two agree to 10^{-12}.

Many of the results are formalized in Lean 4, in the folder `kagey-problems/problem131/lean/PaperA`. This includes Theorem 3.4 and Proposition 3.8 for all $N \ge 2$, and all four parts of Corollary 3.7, with explicit constants in place of the error terms of parts (b) and (c). For $N \le 174$ the Lean kernel checks the same certificates as Step 4 of the proof of Theorem 3.4, with rational bounds on ζ_N that Lean proves itself. For part (d) of the corollary, Lean proves the rational bounds at $N = 2774$ and the extension to every larger N. For Proposition 3.8 and $N \le 174$ it checks rational certificates of its own, built on the same brackets. Only the standard axioms are used. The following are not formalized. In Section 4, Lemma 4.6, Theorem 4.7 beyond its leading term, the limit statements of Theorems 4.8 and 4.9 (for the first, Lean checks algebra over $\mathbb{Q}$ assuming the coefficient equations, and for the second it proves the bound (30)), and Corollary 4.10. In Section 5, Theorem 5.2 (even its leading term), Theorem 5.3, Corollary 5.5 and Proposition 5.7. Also not formalized are the remarks and the numbers in the tables. The file `MAP.md` in that folder gives the Lean theorem for each result, or says that there is none.

7 Discussion

The crossing that Stepanyan et al. [1] read as a thermal transition of the finite chain compares one microstate, the all-up configuration, with one magnetization of a broad band. At the crossing the band has width of order $N/\sqrt{\log N}$ in the magnetization, its free energy relative to full order is the parabola of Remark 5.4, and it carries about $\log N$ domain walls (Proposition 5.7). So the energy gap between the two compared states grows like $\log N$, not like N, and the crossing sharpens only like $1/\log N$ in the coupling. In this sense it is a finite-size crossover between order and disorder rather than a sharpening transition, in line with the absence of transition signals in the microcanonical entropy of finite chains [23]. Its location is set by the balance of one wall-free state against the entropy of about $\log N$ walls, which is where the $\log\log N$ term comes from. The boundary shift by $\log 2$ comes from the parity of the number of walls on a ring.

The bounds behind these statements hold at every N and every coupling. Theorem 4.3 bounds the ratio of the partition function at magnetization $a - b$ to the one at full magnetization, so it gives the fixed-magnetization partition function of the free chain, as used in [13, 15], with an explicit relative error, also when the mean number of domain walls grows with N. The approach extends to the d-state Potts chain [6]. A magnetic field would change the weights of the two letters, and we do not treat it here.

Acknowledgments. I thank Peter Kagey, whose Open Problem Collection [24] asks for this crossing for a Galton board whose ball keeps its direction with probability p. I used AI tools in this work, including Claude Opus 5.5 (Anthropic, through Claude Code, in September and October 2026) and GPT-6 Astra (OpenAI). Except for the proofs of Theorem 3.4, Corollary 5.5,

Proposition 5.7 and Remarks 5.4, 5.6 and 5.8, I wrote the proofs and derivations of this paper by hand, without AI assistance, before any revision. I then used Claude Opus 5.5 to check everything I had written for typos and for statements that were technically wrong. It found a fair number of gaps, which I fixed, along with occasional wrong references and unclear wording. I decided every revision and went over each one by hand. The model also drafted rewordings of three passages, which I reviewed and edited. I had conjectured a bracket of the same kind for the crossing at every N, but it had gaps. Near the end of the work the model proposed an improved conjecture, which is Theorem 3.4 as stated here, together with a proof that uses the Bessel bounds of Lemma 3.6. I rechecked and rederived its proof by hand. In a later revision, which recast the introduction in terms of the Ising chain, the model proposed Corollary 5.5 and the comparison in Table 2, and wrote their proofs and the script that computes the table. I checked both by hand. I also checked the proofs, except those of Corollary 5.5, with GPT-6 Astra, which found nothing beyond the earlier check. In the revision for this journal the model proposed Proposition 5.7, Remarks 5.4, 5.6 and 5.8 and Section 7, and wrote their proofs. It also drafted the comparisons with the literature added in that revision and the revised figures and their scripts.

The Lean 4 formalization in `kagey-problems/problem131/lean/PaperA`, including its later extensions, was generated by Claude Opus 5.5, then revised and checked by hand against the corresponding results (see `MAP.md` in that folder). Parts of the Python verification scripts were also written by the model, which cross-checked them against each other. Lastly, I used it for literature search, namely to find and verify sources beyond the ones I had first checked on my own.

I later used Codex (OpenAI) to audit the paper and its sources, implement corrections and add outward interval checks to the Python certificates. The new Python checks were tested against exact rational calculations.

All mathematical claims, proofs and computations in this paper are my responsibility.

Data availability statement

The code and data are in the folder `kagey-problems/problem131` of https://github.com/apemm/Kagey-Problems. The arXiv version is recorded at the release `paperA-arxiv-v3`. The current revision is on the main branch in the same folder. To rerun the checks, run `python -B verify_all.py --paper a` in that folder. The file `SUPPLEMENT.md` says what each script does, and `paperA/COMPUTATION.md` lists the commands and dependencies. The certificates of Theorem 3.4 are listed in `paperA/data/every_n_certificates.csv`, and `paperA/verify_ising_coupling.py` prints Table 2. Any number in Sections 3 and 5 that no script prints is listed in `paperA/data/quoted_values.txt`.

Conflict of interest

The author declares no conflict of interest.

A The crossing at every N

This appendix proves Lemma 3.6, the analytic half of Theorem 3.4, the identity behind the certificates and Proposition 3.8. The letters Y, Y', $\bar{w}$, $\bar{w}'$, r, r', t, t', τ, κ, m, B, D, β, Δ and δ are local to this appendix (each is defined where it first shows up). We round every decimal constant in the safe direction, namely so that the inequality it enters stays true.

A.1 Proof of Lemma 3.6

For integer n, $I_n(z) = \frac{1}{\pi}\int_0^\pi e^{z\cos t}\cos(nt)\,dt$ [8, Eq. 10.32.3], so $I_0(z)+I_1(z) = \frac{1}{\pi}\int_0^\pi e^{z\cos t}(1+\cos t)\,dt$. Substitute $s = 1-\cos t$, so that $\sin t = \sqrt{s(2-s)}$ and $1+\cos t = 2-s$, and then $s = v/z$. This gives

$$\mathcal{E}(z) = \frac{1}{\sqrt{\pi}}\int_0^{2z} e^{-v}v^{-1/2}\sqrt{1-\frac{v}{2z}}\,dv. \tag{40}$$

On $[0,1]$ we have $\sqrt{1-y} = 1-\sum_{k\ge1}\lambda_k y^k$ with every $\lambda_k > 0$, $\lambda_1 = \frac12$ and $\lambda_2 = \frac18$, and $\sum_k \lambda_k = 1$ by monotone convergence as $y\uparrow 1$. So $\sum_{k\ge3}\lambda_k = \frac38$, and $y^k \le y^3$ gives $\sum_{k\ge3}\lambda_k y^k \le \frac38 y^3$. Hence, for $0\le y\le 1$,

$$g_-(y) := 1-\frac{y}{2}-\frac{y^2}{8}-\frac{3y^3}{8} \le \sqrt{1-y} \le 1-\frac{y}{2}-\frac{y^2}{8} \le g_+(y) := 1-\frac{y}{2}-\frac{y^2}{8}+\frac{y^3}{8}.$$

The cubic g_- decreases and $g_-(1) = 0$, so $g_- \le 0$ on $[1,\infty)$. Also $8g_+(y) = (y-2)^2(y+1)+2(y-1)^2+2 > 0$ for $y \ge -1$. Put $y = v/(2z)$ in (40), replace $\sqrt{1-y}$ by $g_\pm$ and extend the integral to ∞. The added piece is at most 0 with g_- and at least 0 with g_+, so the lower bound survives with g_- and the upper with g_+. Since $\frac{1}{\sqrt{\pi}}\int_0^\infty e^{-v}v^{k-1/2}\,dv = \Gamma(k+\frac12)/\Gamma(\frac12)$ equals $1, \frac12, \frac34, \frac{15}{8}$ for $k = 0,1,2,3$, the two bounds integrate to the two sides of the lemma. The bound $\mathcal{E}\le 1$ follows in the same way from $\sqrt{1-y}\le 1$. Finally $-\frac{3}{128z^2}+\frac{15}{512z^3} \le 0$ exactly when $z \ge \frac54$.

A.2 Proof of Theorem 3.4 for $N \ge 175$

Step 1. Bounds that hold for every $\zeta \ge 4$. Fix $N \ge 175$ and write $\zeta = \zeta_N$, $Y = \zeta + \frac{1}{8\zeta}$, $r = Y/N$ and $n(\zeta) = \sqrt{8\zeta/\pi}\,e^\zeta$. Then $n(\zeta_N) = N$ by (8), and $\zeta \ge 4$ since $n(4) < 175$. For $k = 1,2,3$ the function $Y^k/n(\zeta)$ decreases on $[4,\infty)$, because its logarithmic derivative is $k(1-\frac{1}{8\zeta^2})/Y - 1 - \frac{1}{2\zeta} < \frac{k}{Y} - 1 < 0$. The same holds for Y^4/n, since $Y > 4$, and for ζ/n and ζ^2/n. The values at $\zeta = 4$ give, for every $\zeta \ge 4$,

$$r \le 0.023135, \quad \frac{Y^2}{N} \le 0.093262, \quad \frac{Y^3}{N} \le 0.375960, \quad \frac{Y^4}{N} \le 1.51559, \quad \frac{\zeta(\zeta+1)}{N} \le 0.114777.$$

Step 2. The upper bound. Let $\bar w = Y/(1-r)$, so that $N\bar w/(N+\bar w) = Y$, and let $\delta = \bar w(\bar w+2)/(2N)$. By Lemma 3.5(b) and (9) it suffices to show that $\mathcal{G} := \psi(\bar w) - \psi(\zeta) + \log\mathcal{E}(\bar w) + \log(1-\delta) > 0$. We bound the three parts from below. First, $\psi(\bar w)-\psi(\zeta) = (\bar w - Y) + (Y-\zeta) + \frac12\log\frac{Y}{\zeta} + \frac12\log\frac{\bar w}{Y}$, where $\bar w - Y = \frac{Yr}{1-r}$, $Y-\zeta = \frac{1}{8\zeta}$, $\log\frac{Y}{\zeta} = \log(1+\frac{1}{8\zeta^2}) \ge \frac{1}{8\zeta^2} - \frac{1}{128\zeta^4}$ and $\log\frac{\bar w}{Y} = -\log(1-r) \ge r$. Second, Lemma 3.6 gives $\mathcal{E}(\bar w) \ge 1-t$ with $t = \frac{1}{8\bar w} + \frac{3}{128\bar w^2} + \frac{45}{512\bar w^3}$. Since $\bar w \ge Y \ge \zeta$ and $\frac{1}{8Y} = \frac{1}{8\zeta} - \frac{1}{8\zeta(8\zeta^2+1)} \le \frac{1}{8\zeta} - \frac{1}{64\zeta^3} + \frac{1}{512\zeta^5}$, we get $t \le \tau := \frac{1}{8\zeta} + \frac{3}{128\zeta^2} + \frac{37}{512\zeta^3} + \frac{1}{512\zeta^5}$ and $\log\mathcal{E}(\bar w) \ge \log(1-\tau) \ge -\tau - \frac{\tau^2}{2(1-\tau)}$. Third, $\delta = \frac{Yr}{2(1-r)^2} + \frac{r}{1-r}$ and $\log(1-\delta) \ge -\delta - \frac{\delta^2}{2(1-\delta)}$. Adding the three parts gives $\mathcal{G} \ge B + D$, where B collects the terms in ζ alone and D the terms with a factor r,

$$B = \frac{5}{128\zeta^2} - \frac{37}{512\zeta^3} - \frac{1}{256\zeta^4} - \frac{1}{512\zeta^5} - \frac{\tau^2}{2(1-\tau)}, \qquad D = \frac{r}{2(1-r)}\Big(\frac{Y(1-2r)}{1-r} - 1 - r\Big) - \frac{\delta^2}{2(1-\delta)}.$$

Both $\zeta\tau$ and τ decrease in ζ, so $\zeta^2 B$ increases. At $\zeta = 4$ we have $\zeta\tau < 0.135384$, $\tau < 0.033846$ and $\zeta^2 B > 0.011236$. Hence $B > 0$. In D the bracket is at least $4.03125(1-0.023683) - 1.023135 > 2.912$, so the first term of D is at least $1.456\,r$. Write $\delta = rm$ with $m = \frac{Y}{2(1-r)^2} + \frac{1}{1-r}$. Then $\delta \le 0.07255$

and $rm^2 = \frac{Y^3}{4N(1-r)^4} + \frac{Y^2}{N(1-r)^3} + \frac{r}{(1-r)^2} \le 0.2276$, so $\frac{\delta^2}{2(1-\delta)} = r \cdot \frac{rm^2}{2(1-\delta)} \le 0.1228\, r$. Hence $D \ge 1.33\, r > 0$. So $\mathcal{G} > 0$ and $z_N < Y$.

Step 3. The lower bound. Let $\varepsilon = \varepsilon_N$, $Y' = Y - \varepsilon$, $r' = Y'/N$ and $\bar{w}' = Y'/(1-r')$, so that $N\bar{w}'/(N+\bar{w}') = Y'$. By Step 1, $\varepsilon \le 0.114777 + \frac{1}{256} < 0.1187$, so $Y' \ge \zeta - 0.1187 \ge 3.88$ and $\bar{w}' \ge Y'$. By Lemma 3.5(a) and (9) it suffices to show that $\mathcal{G}' := \psi(\bar{w}') - \psi(\zeta) + \log \mathcal{E}(\bar{w}') < 0$. We bound the two parts from above. First, $\psi(\bar{w}') - \psi(\zeta) = (Y' - \zeta) + \frac{Y'r'}{1-r'} + \frac{1}{2}\log\frac{Y'}{\zeta} - \frac{1}{2}\log(1-r')$, where $Y' - \zeta = \frac{1}{8\zeta} - \varepsilon$, $\frac{1}{2}\log\frac{Y'}{\zeta} \le \frac{1}{2}(\frac{Y'}{\zeta} - 1) = \frac{1}{16\zeta^2} - \frac{\varepsilon}{2\zeta}$ and $-\frac{1}{2}\log(1-r') \le \frac{r'}{2(1-r')}$. The two terms in r' add up to $\frac{Y'^2 + Y'/2}{N(1-r')}$. Second, Lemma 3.6 gives $\mathcal{E}(\bar{w}') \le 1 - t'$ with $t' = \frac{1}{8\bar{w}'} + \frac{3}{128\bar{w}'^2} - \frac{15}{512\bar{w}'^3} \in (0,1)$, so $\log \mathcal{E}(\bar{w}') \le -t'$. Here $\frac{1}{8\bar{w}'} = \frac{1}{8Y'} - \frac{1}{8N}$. Write $Y' = \zeta(1+\kappa)$ with $\kappa = \frac{1}{8\zeta^2} - \frac{\varepsilon}{\zeta}$. Then $\frac{1}{Y'} \ge \frac{1-\kappa}{\zeta}$, so $-\frac{1}{8\bar{w}'} \le -\frac{1}{8\zeta} + \frac{1}{64\zeta^3} - \frac{\varepsilon}{8\zeta^2} + \frac{1}{8N}$. Next, $x \mapsto x/(1-x/N)$ increases and $Y' \le Y$, so $\bar{w}' \le Y/(1-r)$ and $\frac{1}{\bar{w}'^2} \ge \frac{(1-r)^2}{\zeta^2(1+\frac{1}{8\zeta^2})^2} \ge \frac{(1-2r)(1-\frac{1}{4\zeta^2})}{\zeta^2}$. Finally $\frac{15}{512\bar{w}'^3} \le \frac{15}{512Y'^3}$. Adding, and using $\varepsilon(1+\frac{1}{2\zeta}) = \frac{\zeta^2 + \frac{3}{2}\zeta + \frac{1}{2}}{N} + \frac{1}{16\zeta^2} + \frac{1}{32\zeta^3}$, we get $\mathcal{G}' \le \beta + \Delta - \frac{\varepsilon}{8\zeta^2}$ with

$$\beta = -\frac{3}{128\zeta^2}(1-2r)\Big(1 - \frac{1}{4\zeta^2}\Big) + \frac{15}{512Y'^3} - \frac{1}{64\zeta^3}, \qquad \Delta = \frac{1}{N}\Big(\frac{Y'^2 + Y'/2}{1-r'} + \frac{1}{8} - \zeta^2 - \frac{3}{2}\zeta - \frac{1}{2}\Big).$$

For $\beta < 0$ it suffices that $\zeta^2/Y'^3 \le \frac{4}{5}(1-2r)(1-\frac{1}{4\zeta^2})$. The left side is at most $\zeta^2/(\zeta - 0.12)^3 \le 16/3.88^3 < 0.274$, since this function decreases for $\zeta > 0.12$ (its logarithmic derivative is $\frac{2}{\zeta} - \frac{3}{\zeta - 0.12} < 0$ there). The right side is at least $0.8 \cdot 0.9537 \cdot 0.9843 > 0.75$. For $\Delta < 0$, note that $Y' \le Y$ and $r' \le r$, so

$$\frac{Y'^2 + Y'/2}{1-r'} \le Y^2 + \frac{Y}{2} + \frac{Y^3/N + Y^2/(2N)}{1-r} \le \zeta^2 + \frac{\zeta}{2} + 0.2667 + 0.4326,$$

using $Y^2 + \frac{Y}{2} = \zeta^2 + \frac{\zeta}{2} + \frac{1}{4} + \frac{1}{16\zeta} + \frac{1}{64\zeta^2}$. This is less than $\zeta^2 + \frac{3}{2}\zeta + \frac{3}{8}$ because $\zeta \ge 4$. Hence $\mathcal{G}' < 0$ and $z_N > Y'$.

The constants were recomputed in 40-digit arithmetic, and as a check outside the proof we evaluated $\zeta^2 B$, D, β and Δ on $\zeta \in [4, 60]$ with step 0.001.

A.3 The certificates

Let $c, d > 0$ be integers. Multiplying (6) by d^{N-1} gives

$$d^{N-1} S_N(c/d) = \sum_j W(N, m_N, j)\, c^j d^{N-1-j},$$

an integer. By Theorem 2.1 it is the sum of $d^{\mathrm{rep}(w)} c^{\mathrm{chg}(w)}$ over the words w in bin m_N, that is, the numerator $n_N(m_N)$ of Remark 2.3 with $(\alpha, \beta) = (d, c)$. The recursion (3) with the weights d, c in place of p, q and initial values 1 computes the same sums. So $S_N(c/d) < 1$ exactly when this integer is less than d^{N-1}. This is the identity used in Step 4 of the proof of Theorem 3.4.

A.4 Proof of Proposition 3.8

Step 1. An exact formula for R_N. Write $w = w_* = Nu_N$, $z = z_N$, $\Lambda_N = 2\Phi(w)/N$ and $\delta = w(w+2)/(2N)$. At $w = w_*$ the display in the proof of Lemma 3.5 gives $1 \le \Lambda_N \le 1/(1-\delta)$

when $\delta < 1$, and (9) gives $\psi(w) + \log \mathcal{E}(w) = L + c_0 + \log \Lambda_N$. Since $w - z = wz/N$, $w/z = 1 + w/N$ and $\frac{1}{8z} = \frac{1}{8w} + \frac{1}{8N}$, the remainder $R_N = \psi(z) - L - c_0 - \frac{1}{8z}$ is

$$R_N = \log \Lambda_N - \Big(\log \mathcal{E}(w) + \frac{1}{8w}\Big) - \frac{1}{8N} - \frac{wz}{N} - \frac{1}{2}\log\Big(1 + \frac{w}{N}\Big). \tag{41}$$

Step 2. The lower bound for $N \geq 175$. By the proof of Theorem 3.4, $3.88 \leq z \leq w \leq Y/(1-r)$. Note that $\log \Lambda_N \geq 0$ and $-\log \mathcal{E}(w) \geq -\log(1 - \frac{1}{8w}) \geq \frac{1}{8w}$ by Lemma 3.6, since $w \geq \frac{5}{4}$. So $R_N \geq -\frac{wz + w/2 + 1/8}{N} = -\frac{z^2 + z/2}{N - z} - \frac{1}{8N}$. This is at least $-\frac{z^2+z}{N}$ as soon as $N \geq \frac{2z^2(z+1)}{z - 1/4}$. The right side increases in z and $z \leq Y$, so it suffices that $Y^2/N \leq \frac{Y - 1/4}{2(Y+1)}$. The left side is at most 0.0933 and the right side at least 0.3757.

Step 3. The upper bound for $N \geq 175$. Let $t = \frac{1}{8w} + \frac{3}{128w^2} + \frac{45}{512w^3} \leq \frac{1.09504}{8w}$, which holds as $w \geq 3.88$. Lemma 3.6 and $-\log(1-t) \leq t + \frac{t^2}{2(1-t)}$ give

$$-\log \mathcal{E}(w) - \frac{1}{8w} \leq \frac{3}{128w^2} + \frac{45}{512w^3} + \frac{t^2}{2(1-t)} \leq \frac{0.0559}{w^2} \leq \frac{0.0559}{z^2}.$$

Using $\log \Lambda_N \leq \delta + \frac{\delta^2}{2(1-\delta)}$, $\log(1+x) \geq x - \frac{x^2}{2}$ and $wz \geq w^2 - w^3/N$, the rest of (41) is at most $-\frac{(w-1/2)^2}{2N} + \frac{w^3 + w^2/4}{N^2} + \frac{\delta^2}{2(1-\delta)}$. By the bounds on r and Y^k/N in Step 1 of the previous proof, with $w \leq Y/(1-r)$, we have $\frac{w^3 + w^2/4}{N} \leq 0.4278$ and $N\frac{\delta^2}{2(1-\delta)} = \frac{(w^4 + 4w^3 + 4w^2)/(4N)}{2(1-\delta)} \leq 0.4945$, while $\frac{(w-1/2)^2}{2} \geq 5.71$. So the rest is negative and $R_N \leq 0.0559/z^2 < \frac{1}{16z^2}$.

Step 4. The form in L for $N \geq 175$. We have $z < Y = L + c_0 - \frac{1}{2}\log\zeta + \frac{1}{8\zeta} < L$. Also $z > \zeta - 0.1187 \geq \nu_N - 0.1187 \geq L/2$, because $L - \log L \geq 1.18$ for $L \geq \log 175$. Then $\frac{1}{16z^2} \leq \frac{1}{4L^2}$ and $\frac{z^2+z}{N} \leq \frac{L^2+L}{N}$.

Step 5. Short chains. Here $N \leq 174$. The function $\phi(z) = \psi(z) - L - c_0 - \frac{1}{8z}$ increases in z, and $R_N = \phi(z_N)$, while $\frac{1}{16z^2}$ and $-\frac{z^2+z}{N}$ decrease. So it suffices to check $\phi(z(c'/d)) \leq \frac{1}{16z(c'/d)^2}$ and $\phi(z(c/d)) \geq -\frac{z(c/d)^2 + z(c/d)}{N}$ with the certificates, together with $z(c/d) \geq L/2$ and, for $N \geq 3$, $z(c'/d) \leq L$. The outward interval calculation described in Step 4 of the proof of Theorem 3.4 verifies every comparison. The gaps exceed 0.037 (upper) and 0.081 (lower), with the smallest at $N = 174$. For $N = 2$ we have $S_2(u) = 2u$, so $u_2 = \frac{1}{2}$ and $z_2 = \frac{2}{3}$, and $\frac{2}{3} < \log 2$ is equivalent to $e^2 < 8$, which holds since $e^2 < 2.72^2 = 7.3984$.

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